

Howen VSS Web API V2.6.0 R1

Version Log

Version	Description	Date
V1.0	Initial Document	2020.7.29
V1.1	- Add description of real-time message subscription interface - Added web video integration interface description	2020.9.21
V1.2	- Add device supports specifying deviceId; - Add the definition of video request flv http url in Chapter 4; - Add the error code 10043 - Add 2.11 device control web api - Add web player sdk description	2020.11.27
V1.5R1	- Add the description of the location object in the alarm message in the device event message push - Add DMS/ADAS alarm subtype description - Added description of alarm details parameters	2021.2.6
V1.5R2	- Add the description of fast-forward and double-speed parameters to the device video playback URL (supported version: in VSS1.2 and newer) - Add 2.8: Query device GPS and detailed status data by 'page' (supported version: VSS1.5 and newer) - Add 2.10: Query historical alarm details by 'page' (supported version: VSS1.5 and newer) - Chapter 4 add: add the web URL of the server alarm video playback. (supported version: VSS1.2 and newer)	2021.3.5
V1.5R3	- Add: Integration guideline (Chapter 1) - Add: the cookie parameter sent in seconds in the api login interface (3.1) - Add: descriptions of cookie related fields has been added in the interface of 3.8/3.10 - Modify: Equipment alarm event type, upgrade the DMS/ADAS sub-type to the first-level type, the definition of the corresponding first-level type has been added in the equipment event code table (7.9)	2021.3.19
V1.6R1	- Modify: The interface address and request method for querying device GPS and detailed status data by page have changed (the original address and request method are still retained) - Add: more alarm details parameters in "Query historical alarms by page" - Add: interface of obtain device list - Add: interface of download videos in device side - Add: device intercom integration page - Modify: description of location in video file search	2021.4.6
V1.6R2	- Add: the flow chart of Websocket signaling interaction in the [integration guideline chapter] - Modify: Classify the new/modify/delete/device list query interfaces and put them in the device information management interface - Add the interface to obtain the online status of multiple devices - Add the interface for querying the current GPS status of the device - Add the interface for querying alarm evidence - Modify the definition of websocket interface: the login interface does not contain subscription operations, and a separate subscription instruction is added; - Web page embedding method in video integration, support for mobile browser	2021.4.16
V1.7R1	3.11 Video file query interface, add a new function: search center video files 5.5 Server alarm small video playback web page URL, added: playback server recording clip	2021.7.23
V1.8.1R1	- Add: device control command transparent transmission interface (supported by vss1.7), supporting all [device control commands] to be issued to the device, including PTZ, new text distribution and other commands (3.14) - Add: passenger flow statistics report interface (3.16) - Add: driver report interface (3.17)	2021.12.09
V1.8.1R2	3.15 modify the interface request method of web video download	2022.6.22
V1.8.5R1	- Added electronic fence related interfaces - Add more alarm event types	2022.7.19
V1.8.5R2	- Added paraJson detailed parameter parsing of alarm message - Add device ID to query trip report	2022.07.30
V1.8.5R3	- Add parameters in paraJson - Add types in events	2022.8.18
V1.8.5R4	8.10 Add vt in geofence	2022.9.21
V1.8.5R5	8.11 [Alarm details]. Refer to H-protocol document	2022.9.22
V1.8.6R1	3.15 Add two new status to the web video download task state interface (VSS1.8.5 and later versions) 4.4, 4.6 Add: OBD data description and Bluetooth description to Websocket status and alarm messages (VSS1.8.6) 3.3.1 Add: node fields in the new interface of the device (VSS1.8.5 and later versions) 8.1 Add: new http interface error codes (VSS1.8.5 and later versions)	2023.1.29
V1.8.6R2	3.15.1 Add: [video download task] interface, add [task creation time] in the returned result 3.11 Modify: ChannelList field is mandatory 5.4 Add: more descriptions of the speed parameter 3.15.4 Deleted the description of [downUrl]	2023.05.15

	3.13 Device control command add [sending text] and [mileage reset] (v1.8.5 already supported)	
V1.8.6R3	3.9 Updated the meaning of alarmState 3.18 The GEO fence interface is designed and opened for the T project. It is an application-level integration. It is not recommended for other customers to integrate (Note: other interfaces are mainly data-level interfaces) 3.19 Added parameter configuration interface description 5.2 Modified the intercom url format integration - Chapter 7 Delete the flv integration solution and abandon the flv solution 8.9 Added description: For event codes, please refer to the device event specification table of the product function matrix table 8.10 Added description: For codes, please refer to the device event specification table of the product function matrix table	2023.11.08
V1.8.6R4	3.15.4 The web video download task list query interface, adds a guid description to the returned results.	2023.12.28
V2.2.0R1	3.20 Add driver management interface (this interface is applicable to vss v1.8.6 or above)	2024.2.5
V2.2.0R2	3.11 Get whether there is a recording under the date 3.15 New: web video download task, supports downloading of separate GPS data 3.21 New: Related interfaces for language file management 3.22 New: Interface for traffic consumption recording 6 New: Player SDK interface for custom UI 8.11: Add: contents of alarm details parameter analyze 8.9.1: Add: Link of event type code	2024.4.22
V2.3.1.R1	- Add the data for fields added in the latest VSS version. - Added the recently introduced error codes in the latest VSS version	2025.1.08
V2.3.1.R1	Add: instructions for calling high-load interfaces, including GPS tracks, historical alarms, and alarm evidence search	2025.2.18
V2.6.0.R1	4.15.2 Add: web download status	2025.6.26

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1 Fundamental Introduction

1.1 API Types

1. http communication interface

The following functions can be achieved through this type of interface:

1. Interface login. Before using all types of APIs, you need to get a license token through this login interface.
2. Obtain historical business data of the VSS platform, such as GPS, alarm events, video records, Video evidence, current state of the vehicle etc.;
3. Synchronize basic operating data, such as synchronizing MDVR device information.
4. Remotely issue control commands to MDVR equipment

2. Websocket communication interface

Through this type of interface, you can subscribe to VSS real-time business data, such as GPS, alarms, online and offline. When VSS receives information such as GPS/alarms reported by MDVR, it can be distributed to subscribers immediately.

3. Javascript SDK

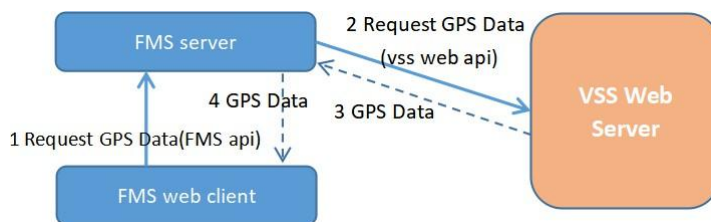
You can integrate the javascript SDK provided by VSS in your web client to, quickly realize video preview and video playback on MDVR and VSS servers.

4. Web embed

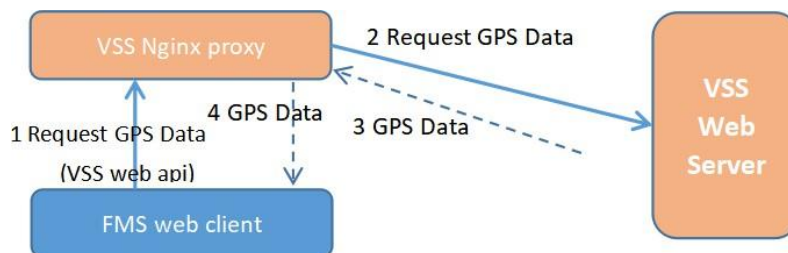
You can directly use the iframe method to embed the VSS web player page in the web page, to quickly realize the video preview and video playback functions on the MDVR and VSS servers.

1.2 Http API Integration Architecture

Architecture 1: Forward data to the FMS client through the FMS backend

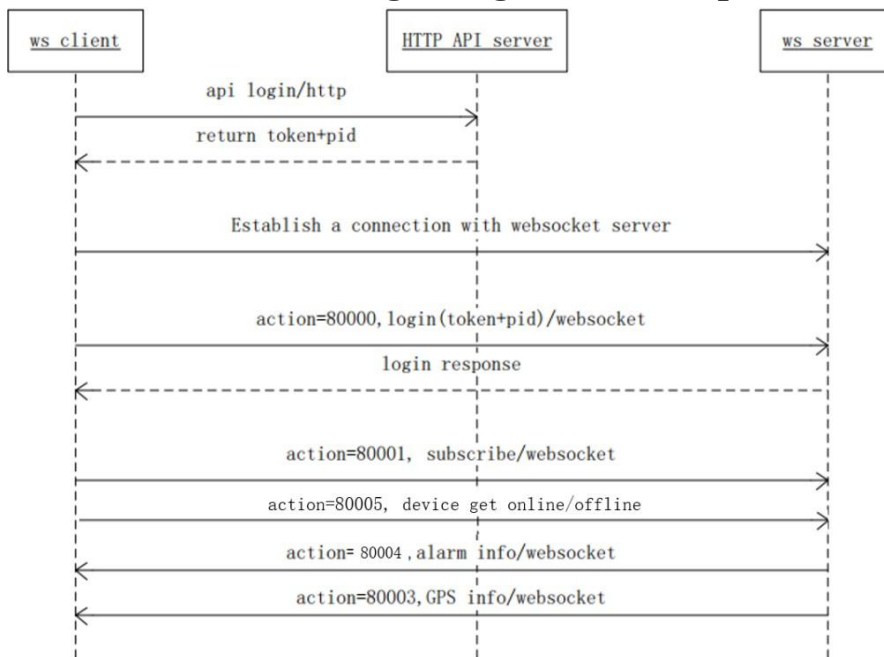


Architecture 2: FMS client directly requests data from VSS (cross-domain)



Note: With this solution, because the browser and VSS Web Server prohibit cross-domain access by default, you need to contact VSS technical support to open the Nginx proxy service. Note: Websocket doesn't have this issue

1.3 Websocket API Signaling interaction process



1. Before the Websocket client connects to the Websocket server, it must obtain the session token and pid parameters through the login interface in the http api.
2. Initiate a connection to the Websocket server and establish a Websocket communication link
3. Client initiates a login instruction to the web server, action=80000
4. Client initiates a message subscription instruction to the web server, action=80001, after completing the message subscription, it waits for the server to actively push GPS/alarm and other messages to the client

2 Transmission Protocol

This interface includes two transmission protocol: HTTP Post and WebSocket.

2.1 HTTP Transmission Protocol

The interface service: HTTP is provided by the VSS platform and follows the HTTP standard protocol (RFC2626)

The default server port: 9966

2.1.1 Get/Post Request Method

1. The Post method support two different data formats.

(1) For sheet submission, shall be **"application/x-www-form-urlencoded"**.

Within "HTTP Message Body": para1=value1¶2=value2

(2) JSON

Format in "HTTP Message Body" is as below:

```
{
  "token": "2a545efadfpqoweirsdfs",
  "para1": "value1",
  "para2": "value2"
}
```

Most http interfaces in this document use post commands

2. For Get method, the request parameters are involved in the URL, as below:

http://vss.ptdigital.co.id/vss/login.action?callback=_getDataFromVss&token=2a545efadfpqoweirsdfs¶1=value1¶2=value2

2.1.2 Common Request Parameters

The request parameters include "Common parameters" and "Private parameters"

Common parameters show as below

Name	Type	Max length	Necessary	Details
token	string	64	yes	Interface access token, all <i>*non-login interface*</i> must carry this parameter, and the value of this parameter is returned after the interface is successfully logged in.
callback	string	32	no	The parameters must be carried, if it is the browser cross-domain access. It will be return the relevant JavaScript execution code, after VSS detects this parameter. (Callback value can only include the letter, number, underline and etc.)

2.1.3 Interface response

JSON type

For the domain access, the JSON data returned in the Message Body. MIME type: **Content-type:**

application/json; charset=UTF-8

JSON data follow the "RFC 4627"

2.1.4 Common response parameters

Name	Type	Max length	Necessary	Details
status	Int		yes	Interfaces return to interfaces. For details, please refer to Return Code Table
data	Json Object		yes	The returned business data object. For details, it will depend on the definition of each interface.

Common Format as below:

```
{
  "status": 10000, "msg": "Success",
  "data": {
    "para1": "value1",
    "para2": "value2"
  }
}
```

2.1.5 Character Set

Both the request data and the returned data adopts UTF-8

2.1.6 HTTPS

The https server default port is 443 port, and the other request format is the same as http.

2.2 Transfer Protocol: WebSocket

The "WebSocket" service is provided by the VSS platform. ws Port: 36300, wss port :36301 ;

All type of the message subscription and the real-time message pushing interfaces adopt the "WebSocket" protocol (RFC6455). In addition, the transmission data adopts JSON format. (The messages described in this document mainly refer to device GPS message, Alarm message, Online /Offline message and soon)

3 Http Interface Definition

3.1 API Access Login in

Interface address: <https://vss.ptdigital.co.id/vss/user/apiLogin.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameters	Type	Length		Required?	Details	Example Value
username	string			yes	User name from VSS Account	Training_Prime
password	string			yes	Password md5 string from VSS Account	The password is converted into MD5 hash For example : Prime123 converted into MD5 hash is 3947562b11fa6f20d27edffd7e7a8007

JSON format example:

```
{
  "username": "admin", "password": "3947562b11fa6f20d27edffd7e7a8007"
}
```

Request succeed, return data example:

```
{
  "status": 10000,
  "msg": "Success",
  "error": null,
  "data": {
    "token": "65292d4efac748beb1f2f530bd13f99b",
    "pid": "d39744e411f814aa5d50fbcee08c18c6fd3156c2a3ed80fb1ad1221f37874659bd187d350fa949b2ff9dda8436467b1"
  },
  "count": 0
}
```

Descriptions of Return parameters

Parameter	Type	Length	Necessary	Description
token	string		Yes	The http interface token obtained after successful login. Needs to carry this token in subsequent api interfaces. Token validity period: 30 minutes, if there is no interface interaction within 30 minutes, the token will expire
pid	string		Yes	Websocket interface token

Others return data example:

```
{
  "status": 10003, // Status code, refer to the return code table
  "msg": "Server error", //Indication
  "data": null
}
```

3.2 API Login Out

Interface address: <https://vss.ptdigital.co.id/vss/user/apiLogout.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example Value
username	string		yes	User name	admin
token	string		yes	Token	

Request succeed return data example:

```
{
  "status": 10000, // Return status code, refer to return code table
  "msg": "Success", //Return Indication
  "data": null
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server Error", //Indication
  "data": null
}
```

3.3 Interface of Device information management

Device information management interface, used for real-time synchronization and management of device information between the external platform and the VSS platform.

3.3.1 New Equipment

Interface address: <http://vss.ptdigital.co.id/vss/vehicle/apiAddDevice.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example value
deviceID	string	64	yes	Device ID	dev001
deviceName	string	64	yes	Device Name	
deviceType	string		yes	Device Type code. For details, please refer to Device Type code	Hero-ME41-04
channelName	String	256	No	Channel Names Set. Channels need to be separate by semicolons (;)	ch1;ch2;ch3;ch4
fleetId	String	255	No	Fleet ID, if it is empty, means it belongs to default fleet	fleet001 Note: The account must obtain this fleet ID authority
token	String		Yes	Token	
ioName	String		No	io name, each io is separated by a semicolon	io1; io2; io3...; io12
nodeGuid	String		No	Node server number, default value: 0, 0 means the central node	0
chassisNo	String		No	Chassis Number	

Json format example of request:

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "deviceID": "99990001", "deviceName": "90001",
  "deviceType": "Hero-ME41-04",
  "channelName": "ch1;ch2;ch3;ch4",
  "nodeGu": "0"
}
```

Request succeed return data example:

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server error", //Indication
  "data": null
}
```

3.3.2 Modify Equipment

Interface address: <http://vss.ptdigital.co.id/vss/vehicle/apiModifyDevice.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	details	Example value
deviceID	string	64	yes	Device ID	v001
deviceName	string	64	yes	Device Name	
channelName	String	256	no	Channel Names Set. Channels need to be separate by semicolons(;	ch1;ch2;ch3;ch4

deviceType	string		yes	Device Type code. For details, please refer to Device Type code	Hero-ME41-04
token	String		yes	Token	
chassisNo	String		No	Chassis Number	

Json format example of request:

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "deviceID": "99990001",
  "deviceName": "90001",
  "channelName": "ch1;ch2;ch3;ch4",
}
```

Request succeed return data example:

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server error", //Indication
  "data": null
}
```

3.3.3 Delete Equipment

Interface address: <http://vss.ptdigital.co.id/vss/vehicle/apiRemoveDevice.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example Value
deviceID	string	64	yes	Device ID	Fleet001
token	String		yes	Token	

Json format example of request:

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "deviceID": "99990001",
}
```

Request succeed return data example:

```
{
  "status": 10000, "msg": "Success",
  "data": null
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server error", //Indication
  "data": null
}
```

3.3.4 Obtain device list by page

interface address: <http://192.168.1.2:9966/vss/vehicle/findAll.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameters	Type	Length	Necessa ry	Details	Example Value
pageNum	number		Yes	Page number, set to -1 when querying all	2
pageCount	number		Yes	Number per page, set to -1 when querying all	20

isOnline	string		No	Check whether the device is online. 1- means only get online devices 0- means to obtain offline devices. If it is empty, query all devices	1
keyword	string		No	Search for the keyword. Search by device ID or device name	
fleetid	String		No	Fleet ID, multiple fleet IDs are separated by commas	
sort	String		No	asc (ascending sequence), desc (descending sequence)	
sortField	String		No	Sequencing field: deviceName, lastOnlineTime, accessMode (the status of last time online)	

Json format example of request:

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "isOnline": "1",
  "pageCount": "20000",
  "pageNum": "1",
  "keyword": ""
}
```

Returned parameter by page:

Parameters	Type	Necessary	Details	Example Value
totalCount	number	Yes	Total number of records	
pageCount	number	Yes	Number of records per page	
fromCount	number	Yes	The starting record number of the current page	
toCount	number	Yes	The last record number of the current page	
totalNum	number	Yes	total pages	
pageNum	number	Yes	The current page number, starting from 1	

Return the attribute parameter table of each device:

Parameters	Type	Length	Necessary	Details	Example Value
deviceno	number		Yes	Device ID, unique	
devicename	number		Yes	Device name	
devicetype	number		No	Device type	
fleetid	String		Yes	Fleet id	
createtime	String		Yes	Device creat time	
videoencodernumber	number		Yes	Number of video channels of the device	
speakernumber	String		Yes	Number of device intercom channels	
accessmode	number		Yes	The online status of the device. Refer: Device status table	
lastonlinetime	String		Yes	Last online time	
lastofflinetime	String		Yes	Last offline time	
longitude	String		Yes	The longitude of the last positioning	
latitude	String		Yes	The latitude of last positioning	
dayUsed	number		Yes	Data cost today, unit: Byte	
monthUsed	number		Yes	Data cost this month, unit: Byte	
plateno	String			License plate number	
vehicletype	String			Vehicle type	
sim	String			SIM card number	
chassisno	String			Chassis number (VIN)	
installDate	String			Vehicle installation date	
channelname	String			Channel name	

engineno	String			Engine number	
identiyao	String			Vehicle registration number	
remarks	String			Remarks	
modifytime	String			Modification time	
imei	String			IMEI	
imsi	String			IMSI	
phone	String			Phone number	
deviceModel	String			Terminal model	
appVersion	String			Firmware version	
outputnumber	String			Number of output ports	
inputnumber	String			Number of input ports	
icon	String			Icon	
ioname	String			IO name	
nodeGuid	String			Node ID	

The format of the data returned after the request is successful:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "totalCount": 1985,
    "pageCount": 100,
    "fromCount": 1,
    "toCount": 100,
    "totalNum": 20,
    "pageNum": 1,
    "dataList": [
      {
        "guid": "B5A41067-6FB24946-9D4C7E3F-140A115E",
        "deviceno": "00000000",
        "devicename": "00000000",
        "devicetype": "Hero-ME41-02",
        "channelname": "ch1;ch2",
        "sim": "",
        "fleetid": "4A4FE94D-F98A4CDC-B040ED9E-1790838C",
        "remarks": "",
        "createtime": "2021-03-18 11:25",
        "modifytime": "2021-03-18 11:25",
        "videoencodernumber": 2,
        "speakernumber": 1,
        "inputnumber": 2,
        "outputnumber": 2,
        "accessmode": -1,
        "addressinternet": null,
        "userid": null,
        "username": null,
        "lastonlinetime": "2021-04-01 11:49",
        "lastofflinetime": "2021-04-02 00:47",
        "roleid": null,
        "lastStatusJson": "",
        "longitude": 113.946854,
        "latitude": 22.557869,
        "altitude": 0.0,
        "speed": "0",
        "precision": "0",
        "direct": "0",
        "satellites": "0",
        "mode": "1",
        "dtu": "2021-04-02 00:46:57",
        "todaymileage": 0.00,

```

```

        "totalmileage": 0.00,
        "flowLimitDay": -1,
        "flowLimitMonth": -1,
        "flowSettlementDay": 1,
        "dayUsed": 29928,
        "monthUsed": 202239064
    },
}
]
}

```

Examples of returned data in other cases

```

{
    "status": 10003,
    "msg": "Server Error",
    "data": null
}

```

3.4 Get OnlineStatus of single Device

Interface address: <http://vss.ptdigital.co.id/vss/onoffline/queryDeviceStatus.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example Value
deviceID	string	64	yes	Device ID, multiple IDs separated by commas	device001

Json format example of request:

```

{
    "token": "65292d4efac748beb1f2f530bd13f99b", "deviceID": "99990001",
}

```

Private response parameters:

Parameters	Type	Length	Necessary	Details	Example Value
deviceID	string	64	yes	Device ID	Fleet001
queryTime	String		yes	Query Time	
accessMode	Int		yes	Device online status code. For details, please refer to Device status code set	
lastOnlineTime	String		no	Last online time	
lastOfflineTime	String		no	Last offline time	

Request succeed return data example:

```

{
    "status": 10000, "msg": "Success",
    "data": {
        "queryTime": "2020-07-06 12:12:12",
        "accessMode": -1,
        "lastOnlineTime": "2020-07-06 12:12:12",
        "lastOfflineTime": "2020-07-06 14:12:12"
    }
}

```

Others return data example:

```

{
    "status": 10003, "msg": "Server Error",
    "data": null
}

```

3.5 Get the online status of multiple devices

Interface Address: <http://vss.ptdigital.co.id/vss/onoffline/queryMoreDeviceStatus.action>

Request Method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameters	Type	Lens	Necessary	Description	Example
deviceID	string	64	Yes	Device number, separated by commas	99990001,99990002

json format example of Request:

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "deviceID": "99990001,99990002",
}
```

Private request parameters:

Parameters	Type	Lens	Necessary	Description	Example
deviceID	string	64	Yes	Device number	Fleet001
queryTime	String		Yes	Query time	
accessMode	number		Yes	Device online status code Refer: Device status	
lastOnlineTime	String		No	Last online time	
lastOfflineTime	String		No	Last offline time	

Request returned data successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": [
    {
      "queryTime": "2020-07-06 12:12:12",
      "accessMode": -1,
      "lastOnlineTime": "2020-07-06 12:12:12",
      "lastOfflineTime": "2020-07-06 14:12:12"
    }
  ]
}
```

Return of Other conditions:

```
{
  "status": 10003,
  "msg": "server error",
  "data": null
}
```

3.6 Get the current GPS status of the device

This interface is used to obtain the last GPS and status information reported by the device.

Interface Address: <http://vss.ptdigital.co.id/vss/vehicle/getDeviceStatus.action>

Request Method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameters	Type	Lens	Necessary	Description	Example
token	String		Yes	Token returned after successful login	
deviceID	string		Yes	Device number, multiple devices are separated by commas	300330,99990001

json example of Request

```
{
  "token": "f02cfa1cd21138ac89f7f1a32f8339be",
  "deviceID": "4104",
}
```

Private response parameters:

The returned GPS status object parameters are as follows:

Parameters	Type	Lens	Necessary	Description	Example
satellites	number		Yes	Satellite number	
direct	number		Yes	The direction of the	

				vehicle, 0-360 degrees. The true north direction is 0 degrees, increasing clockwise	
speed	number		Yes	Speed, unit km/h	
altitude	number		No	Altitude, unit meter	
longitude	String		Yes	longitude	
latitude	String		Yes	latitude	
precision	number			precision, unit : meter	
createtime	String			Gps time	
accState	number			1-Acc on 0- Acc off	
speed	String			speed	
recordState	number			Channel recording status. Bit0 represents the first channel	
videoMaskState	number			Video occlusion status Bit0 represents the first channel	
videoLostState	number			Video loss status. Bit0 represents the first channel	
ioState	number			io status. Bit0 represents the first channel	
oilVolume	Number			Fuel amount	
tempeAndHumidity	Array Object			Temperature and humidity sensor data array	
isLater	Number			Whether to retransmit data	
reportTime	String			Time to report to the platform	2021-03-02 11:19:38
videoloststateFormatter	String			Video loss status formatted field	
videomaskstateFormatter	String			Video obstruction status formatted field	
iostateFormatter	String			IO status formatted field	
recordstateFormatter	String			Recording status formatted field	
mode	number			Positioning type	
overspeed	number			Speeding status (1 for Yes, 0 for No)	
lowspeed	number			Low speed status (1 for Yes, 0 for No)	
urgency	number			Emergency alarm status (1 for Yes, 0 for No)	
state.Json	String			Status JSON data	
devVoltage	float			Device voltage	
netType	number			Network type	
signalValue	number			Network signal strength	
batVoltage	float			Battery voltage	
driverCardId	String			Driver card number	
tempeAndHumidity	Array			Temperature and humidity value array	
driverName	String			Driver name	

The data format returned by the request is as follows:

```
{
  "status": 10000,
  "msg": "Success",
}
```

```
"data": [
  {
    "guid": "D02B567A-3ACD4C80-A1FC05C6-38F8BB29",
    "deviceguid": "10001",
    "deviceName": "10001",
    "longitude": 113.946609,
    "latitude": 22.5581837,
    "altitude": 52.0,
    "createtime": "2021-03-02 11:19:38",
    "speed": "0",
    "precision": "1",
    "direct": "36",
    "satellites": "3",
    "mode": "1",
    "urgency": "0",
    "stateJson": " {\"mileage\":...}", //all State raw data json
    "overSpeed": "0",
    "accState": "1",
    "lowSpeed": "0",
    "recordState": "255",
    "ioState": "1",
    "oilVolume": 0.0,
    "videoMaskState": "0",
    "videoLostState": "252",
    "tempeAndHumidity": [
      "0",
      "0",
      "0",
      "0",
      "0",
      "0"
    ],
    "isLater": 0,
    "reportTime": "2021-03-02 11:19:39",
  },
],
}
```

Examples of returned data in other cases

```
{
  "status": 10003,
  "msg": "Server Error",
  "data": null
}
```

3.7 Query Device GPS Track by Page

This interface is used to query the historical GPS track of the device



Important Notes:

- (1) This interface consumes a lot of resources (large amount of data, query consumes more disk io), it is recommended not to query too large a time range, and only query a single vehicle track at a time, otherwise it may cause server abnormalities. Please frequently observe the server resource situation to adjust the call
- (2) If you must obtain device historical data at a high frequency, and the server hardware resources are insufficient, please use the websocket Sisi message interface in Chapter 4 to save the GPS and alarm data to your own database, and save the recent high-frequency access data in the memory database.

Interface address: <http://vss.ptdigital.co.id/vss/track/getTrackList.action>

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example
------------	------	--------	-----------	---------	---------

					Value
pageNum	number		yes	Page number. Set to -1 when querying all	2
pageCount	number		yes	Number per page. Set to -1 when querying all	20
deviceID	string		yes	Device ID	300330
beginTime	string		yes	Begin time	2017-01-23 12:12:12
endTime	string		yes	End time	2017-01-23 13:12:12
token	String		yes	Token	

Json format example of request:

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "deviceID": "99990001",
  "pageNum": "1",
  "pageCount": "200",
  "beginTime": "2017-01-23 12:12:12",
  "endTime": "2017-01-23 12:12:12"
}
```

Private response parameters:

Parameters	Type	Length	Necessary	Details	Example Value
deviceID	String			Device ID	
alarmGuid	String			Alarm ID	
deviceName	String			Device name	
type	String			Type (1: gps, 2: alarm)	
location	Object			Gps data	
mileage	Object			Mileage data	
cardNo	String			Driver card number	
driverName	String			Driver name	

Location object parameters show as below:

Parameters	Type	Length	Necessary	Details	Example Value
satellites	number		yes	Number of satellites	
direct	number		yes	Vehicle direction, 0-360 degrees. The north is 0 degree, increasing clockwise.	
speed	number		yes	Speed. Unit: km/h	
altitude	number		yes	Altitude. Unit: meter	
longitude	String		yes	Longitude	
latitude	String		yes	Latitude	
precision	number			Accuracy. Unit: meter	
dtu	String			GPS time	
mode	String			Positioning mode	
alarmValue	String			Alarm type	
devVoltage	Float			Voltage	

Mileage object parameters show as below:

Parameters	Type	Length	Necessary	Details	Example Value
todayDay	number		yes	Total mileage today. Unit: 10 meters	

total	number		yes	Total mileage. Unit: 10 meters	
-------	--------	--	-----	--------------------------------	--

Request succeed return data example:

```
{
  "status": 10000, // Return status code, refer to return code table
  "msg": "Success", //Return indication
  "data": [
    {
      "deviceId": "300331",
      "location": {
        "satellites": "0",
        "direct": "-45",
        "speed": "2",
        "altitude": "221",
        "longitude": "114.113586",
        "latitude": "22.5336876",
        "precision": "50",
        "dtu": "2018-03-13 11:30:06",
        "mode": "1"
      },
      "mileage": {
        "todayDay": "100",
        "total": "20000"
      }
    }
  ]
}
```

Others return data example:

```
{
  "status": 10003,
  "msg": "Server Error",
  "data": null
}
```

3.8 Query device GPS and detailed status data by page

This interface is used to query the historical GPS track and corresponding status data of the device



Important Notes:

- (1) This interface consumes a lot of resources (large amount of data, query consumes more disk io), it is recommended not to query too large a time range, and only query a single vehicle track at a time, otherwise it may cause server abnormalities. Please frequently observe the server resource situation to adjust the call
- (2) If you must obtain device historical data at a high frequency, and the server hardware resources are insufficient, please use the websocket Sisi message interface in Chapter 4 to save the GPS and alarm data to your own database, and save the recent high-frequency access data in the memory database.

Interface address: <http://vss.ptdigital.co.id/vss/track/getApiTrackList.action>

Request method: http POST

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

Parameter	Type	Length	Necessary	Details	Example value
token	String		yes	Token returned after successful login	
pageNum	number		yes	Page number, set to -1 when querying all	2
pageCount	number		yes	Number per page, set to -1 when querying all	20

deviceID	string		yes	Device ID, Multiple devices are separated by commas	300330, 99990001
beginTime	string		yes	Begin Time	2017-01-23 12:12:12
endTime	string		yes	End Time	2017-01-23 13:12:12

Request json example:

```
{
  "token":"f02cfa1cd21138ac89f7f1a32f8339be",
  "deviceID":"4104",
  "beginTime":"2021-03-30 00:00:00",
  "endTime":"2021-04-01 09:00:00",
  "pageNum":"1",
  "pageCount":"200",

```

Private response parameters:

In the returned json data object, the parameters are included in the following table:

Parameter by pages:

Parameter	Type	Necessary	Details	Example value
totalCount	number	yes	Total number of records	
pageCount	number	yes	Number of records per page	
fromCount	number	yes	The starting record number of the current page	
toCount	number	yes	The last record number of the current page	
totalNum	number	yes	Total pages	
pageNum	number	yes	The current page number, starting from 1	

The business data is as follows:

Parameter	Type	Length	Necessary	Details	Example value
satellites	number		yes	Number of satellites	
direct	number		yes	The direction of the vehicle, 0-360 degrees. The north direction is 0 degrees, increasing clockwise	
speed	number		yes	Speed in km/h	
altitude	number		No	Altitude, in meters	
longitude	String		yes	longitude	
latitude	String		yes	latitude	
precision	number			Precision in meters	
createtime	String			Gps time	
accState	number			2-Acc on 1- Acc off	
speed	String			speed	
recordState	number			Channel recording status. Bit0 represents the first channel, and so on	
videoMaskState	number			Video cover status Bit0 represents the first channel, and so on	
videoLostState	number			Video loss status. Bit0 represents the first channel, and so on	
ioState	number			io status. Bit0 represents the first channel, and so on	
oilVolume	Number			Fuel volume	
tempeAndHumidity	Array object			Temperature and humidity sensor data array	
isLater	Number			Whether to retransmit data or not	

reportTime	String			Time of reporting to the platform	2021-03-02 11:19:38
deviceguid	String			Device ID	
deviceName	String			Device name	
videoloststateFormatter	String			Video loss status formatted field	
videomaskstateFormatter	String			Video obstruction status formatted field	
iostateFormatter	String			IO status formatted field	
recordstateFormatter	String			Recording status formatted field	
mode	number			Positioning type	
overspeed	number			Speeding status (1 for Yes, 0 for No)	
lowspeed	number			Low speed status (1 for Yes, 0 for No)	
urgency	number			Emergency alarm status (1 for Yes, 0 for No)	
stateJson	String			Status JSON data	
devVoltage	float			Device voltage	
netType	number			Network type	
signalValue	number			Network signal strength	
batVoltage	float			Battery voltage	
driverCardId	String			Driver card number	
tempeAndHumidity	Array			Temperature and humidity value array	
driverName	String			Driver name	

The format of the data returned after the request is successful is as follows:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "totalCount": 1985,
    "pageCount": 100,
    "fromCount": 1,
    "toCount": 100,
    "totalNum": 20,
    "pageNum": 1,
    "dataList": [
      {
        "guid": "D02B567A-3ACD4C80-A1FC05C6-38F8BB29",
        "deviceguid": "10001",
        "deviceName": "10001",
        "longitude": 113.946609,
        "latitude": 22.5581837,
        "altitude": 52.0,
        "createtime": "2021-03-02 11:19:38",
        "speed": "0",
        "precision": "1",
        "direct": "36",
        "satellites": "3",
        "mode": "1",
        "urgency": "0",
        "stateJson": "{\\\"mileage\\\":...}", //status raw data
        "overSpeed": "0",
        "accState": "1",
        "lowSpeed": "0",
        "recordState": "255",
        "ioState": "1",
        "oilVolume": 0.0,
        "videoMaskState": "0",

```

```

        "videoLostState": "252"
        "tempeAndHumidity": [
            "0",
            "0",
            "0",
            "0",
            "0",
            "0"
        ],
        "isLater": 0,
        "reportTime": "2021-03-02 11:19:39",
    },
]
}

```

Examples of returned data in other cases

```

{
    "status": 10003,
    "msg": "server error",
    "data": null
}

```

3.9 Query History Alarm by Page



Important Notes:

- (1) This interface consumes a lot of resources (large amount of data, query consumes more disk io), it is recommended not to query too large a time range, and only query a single vehicle track at a time, otherwise it may cause server abnormalities. Please frequently observe the server resource situation to adjust the call
- (2) If you must obtain device historical data at a high frequency, and the server hardware resources are insufficient, please use the websocket Sisi message interface in Chapter 4 to save the GPS and alarm data to your own database, and save the recent high-frequency access data in the memory database.

Interface address: <http://vss.ptdigital.co.id/vss/alarm/apiFindAllByTime.action>

Request method: http POST

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessa ry	Details	Example Value
pageNum	number		yes	Page number. Set to -1 when querying all	2
pageCount	number		yes	Number per page. Set to -1 when querying all	20
deviceID	string		yes	Device ID	300330
beginTime	string		yes	Begin time	2017-01-23 12:12:12
endTime	string		yes	End time	2017-01-23 13:12:12
alarmType	String		no	Alarm event type. Blank mean all type. For details, please Refer to Event code	1
driverCardId	String		no	Driver card ID	
token	String		yes	Token	

Json format example of request:

```
{
```



```

"token":"65292d4efac748beb1f2f530bd13f99b",
"deviceID":"99990001", "pageNum":"1",
"pageCount":"200", "beginTime":"2017-01-23 12:12:12",
"alarmType":""
}

```

Private response parameters:

In the return JSON data object, include the Paging parameters and Alarm object array.

Paging parameters show as below:

Parameters	Type	Necessary	Details	Example Value
totalCount	number	yes	Total number of records	
pageCount	number	yes	Number of records per page	
fromCount	number	yes	Starting record number in current page	
toCount	number	yes	End record number in current page	
totalNum	number	yes	Total page number	
pageNum	number	yes	The current page number, starting from 1	

Alarm object parameters show as below:

Parameters	Type	Necessary	Details	Example Value
guid	string		Alarm ID	
deviceID	string		Device ID	
alarmtype	number		Alarm event type. For details, please refer to Alarm Event oCde Table	
alarmvalue	string		Alarm details parameters value: Contains detailed alarm related parameters and alarm subtypes. For parameter details, please refer to the code table section;	{"num":"sd1","st":"0"}。
createtime	string		Time when Alarm event generated	
alarmState	number		Alarm status. 1 means start, 0 means end, 2 means the alarm is ongoing.	
AlarmGps	String		GPS coordinates when the alarm occurs	
Speed	String		The speed of the vehicle at the time the warning occurred	113.946564,22.5582962
Islater	Number		Whether to retransmit data or not	
reportTime	String		Time for reporting to the platform	2021-03-02 11:19:38
paraJson	String		Other parameters related to the event, in json format.	{ "accState":1, "videoLost":"188", "vcc":"11.8000002", "spds":"0", "totalMileage":"55930" }
paraJson.accState	Number		Acc status。 1-on, 0-off	
paraJson.videoLost	Number		Video loss status. Bit0 represents the first channel, and so on, bit0=0 represents no video loss in the first channel, bit0=1 represents video loss in the first channel	
paraJson.vcc	Number		Device voltage	
paraJson.spds	Number		Speed resource 0—GPS speed 1—OBD speed	
paraJson.totalMileage	Number		Total mileage	
paraJson.rpm	Number		RPM of engine	

paraJson.satellites	Number		Quantity of satellites	
paraJson.altitude	Number		Altitude	
deviceName	String		Device name	
drivername	String		Driver name	
driverCardId	String		Driver card number	
rpm	String		OBD speed	
spds	Number		Speed source (0 or null for GPS, 1 for OBD)	
totalMileage	Number		Total mileage	
endTime	String		Alarm end time	
endGps	String		Alarm end GPS	
takeType	String		False alarm flag (1 for false alarm, others for correct alarm)	
alarmTimeLength	String		Alarm duration	
plateNo	String		License plate number	
satellites	Number		Number of satellites	
accState	Number		ACC status	
altitude	Number		Altitude	
videoloststateFor matter	String		Video loss formatted status	
vcc	Float		Device voltage	
reviewType	Number		Cloud re-review type (null: not reviewed, 0: correct alarm, 1: false alarm)	
fenceName	String		Geofence name	
fenceId	String		Geofence ID	
endSpeed	Float		End speed	
endDetail	String		End details	

Request succeed return data example:

```
{
  "status":10000,// Return status code, refer to return code table
  "msg": "Success", //Return
  "data":
  {
    "totalCount": 12, //Total count
    "pageCount":10,//count on each page, -1 mean check all
    "fromCount": 11, //Count starting from current page
    "toCount": 12, // Count ending to current page
    "totalNum": 2, //Total page count
    "pageNum":2, //Current page number
    "dataList":
    [
      {
        "guid": "70814723-2B214A29-98057730-91654774",
        "deviceguid": "333555(333555)",
        "alarmtype": "5",
        "alarmvalue": "io:1; ",
        "createtime": "2021-04-01 09:00:00",
        "alarmGps": "113.946564,22.5582962",
        "speed": "0",
        "takeuptime": null,
        "alarmState": 1,
        "reportTime": "2021-04-01 08:59:55",
        "isLater": 0,

```

```

    "takeupUser": null,
    paraJson: "{\"accState\":1,\"videoLost\":\"0\",\"vcc\":\"11.8000002\",\"spds\":\"0\",\"totalMileage\":\"10380\"}\",
    "takeupMemo": null},
  ]
}
}
}
Others return data example:
{
  "status": 10003,
  "msg": "Server Error",
  "data": null
}

```

3.10 Query Device Online/Offline Record by Page

Interface address: <http://vss.ptdigital.co.id/vss/onoffline/apiFindAll.action>

Request method: http POST

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example value
pageNum	number		yes	Page number, set to -1 when query all	2
pageCount	number		yes	Number of per page, set to -1 when query all	20
deviceID	string		yes	Device ID	300330
beginTime	string		yes	Begin time	2017-01-23 12:12:12
endTime	string		yes	End time	2017-01-23 13:12:12
token	string		yes	Token	

JSON format example of request:

```

{
  "token":"65292d4efac748beb1f2f530bd13f99b",
  "deviceID":"99990001", "pageNum":"1",
  "pageCount":"200", "beginTime":"2017-01-23 12:12",
  "alarmType":""
}

```

Private response parameters:

In the return JSON data object, include the paging parameters and online/offline object array.

Paging parameters show as below:

Parameters	Type	Necessary	Details	Example value
totalCount	number	yes	Total number of records	
pageCount	number	yes	Records number per page	
fromCount	number	yes	Starting record number in current page	
toCount	number	yes	End record number in current page	
totalNum	number	yes	Total number of pages	
pageNum	number	yes	The current page number, starting from 1	

Online/offline object array show as below:

Parameters	Type	Necessary	Details	Example value
guid	string		Record	
deviceID	string		Device ID	
createtime	string		Alarm event	
accessmode	number		Online/Offline status. For details, please refer to Device Online Status Table	

Request succeed return data example:

```

{

```

```
"status":10000,// Return status code, refer to return code table
"msg": "Success", //Return info
"data":
{
  "totalCount": 12, //Total count
  "pageCount":10,//count on each page, -1 mean check all
  "fromCount": 11, //Count starting from current page
  "toCount": 12, // Count ending to current page
  "totalNum": 2, //Total page count
  "pageNum":2, //Current page number
  "dataList":
  [
    {
      "guid":"0E4732CB-59D74CDA-B410C76D-D3268B65", "deviceID":"999908",
      "accessmode":"-1", "createtime":"2020-08-0415:49:56"
    },
  ]
}
}
```

Others return data example:

```
{
  "status": 10003, "msg": "Server Error",
  "data": null
}
```

3.11 Records Searching

Interface address: <http://vss.ptdigital.co.id/vss/record/videoFileSearch.action>

Request method: http POST

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters:

Parameters	Type	Length	Necessary	Details	Example value
deviceID	string	64	yes	Device ID	Fleet001
startTime	String	19	yes	Starting time. If the date format is yyyy-MM-dd, it means querying whether there is a video on the date	2017-12-20 00:00:00
endTime	String	19	yes	End time If the date format is yyyy-MM-dd, it means querying whether there is a video on the date	2017-12-20 23:59:59
channelList	String	32	yes	The recording channel range, separated by semicolons. The channel number starts from 1.	1;2;3
fileType	Int		yes	1-Ordinary video, 2- Alarm video, 3-Normal picture, 4- Alarm picture	1
location	Int		yes	Video storage location. 1-Device 3-Automatic Download Server 4-Alarm small video server (including http small video and platform alarm triggered small video) 5-Server recording files	1
scheme	string		no	Video download address protocol type. It can only be http or https, default http	https
token	string		yes	Token	
stream	Number		no	0 - Substream, 1 - Mainstream, default is Mainstream	

Example of Request JSON format:

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "deviceId": "99990001", "startTime": "2017-12-2000:00:00",
  "endTime": "2017-12-2021:00:00",
  "channelList": "1;2",
  "fileType": "1",
  "location": "1"
}
```

Private response parameters:

In the return JSON array object: Files and array element object parameters show as below:

Parameters	Type	Length	Necessary	Details	Example value
deviceId	string	64	yes	Device ID	Fleet001
channel	Int		yes	Channel number start from 1.	
duration	Int		no	Time. Unit: second	
size	Int		no	Files size. Unit: Byte	
type	Int		yes	1-Ordinary video, 2- Alarm video, 3-Normal picture, 4- Alarm picture	
start	String	20	yes	Document starting time	
stop	String	20	yes	Document end time	
path	String	256	no	Files full path	
alarmType	Number		no	Alarm type. refer to: alarm event code table	
downUrl	String	255	no	Video download address. Currently only supports downloading of files such as small videos and pictures of server-side alarms	
plateNo	String			License plate number	
deviceName	String			Device name	
stream	Number			0 - Substream, 1 - Mainstream, default is Mainstream	
diskName	String			Disk name	

Request succeed return data example:

```
{
  "status": 10000, // Return status code, refer to return code table
  "msg": "Success", //Return indication
  "data": {
    "files": [
      {
        "deviceId": "300330",
        "channel": "1",
        "duration": "296",
        "size": "97991535",
        "type": "1",
        "start": "2017-12-2000:00:00",
        "stop": "2017-12-2000:04:56",
        "path": "/0000_00870000_5_1513728000_1513728296_0_97991535.h264"
      },
      {
        "deviceId": "300330",
        "channel": "2",
        "duration": "296",
        "size": "9943402",
        "type": "1",
        "start": "2017-12-2000:00:00",
        "stop": "2017-12-2000:04:56",
        "path": "/0001_00870200_5_1513728000_1513728296_0_9943402.h264"
      },
      {
        "alarmTyp": 132,

```

3.12 Search Alarm Evidence



New Interface Address:<http://serverIp:9966/vss/record/findEvidences.action>

Old Interface Address: <http://serverIp:9966/vss/record/evidenceToRetrieve.action>

REMARK: The old interface will be deleted in the future. Please switch to the new interface in time.

Request method: http POST

Data format: application/json or application/x-www-form-urlencoded

Private request parameters:

30

				unprocessed)	
reviewType	Number			DMS cloud re-review (null: all, 0: correct alarm, 1: false alarm)	

Json format example of Request:

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "conditionName": "99990001",
  "startTime": "2017-12-20 00:00:00",
  "endTime": "2017-12-20 21:00:00",
  "alarmType": "",
  "scheme": "http"
}
```

Private response parameters:

Return an array of alarm objects, the parameters of alarm objects are as follows:

Parameters	Type	Length	Necessary	Details	Example value
deviceID	string	64	Yes	device ID	Fleet001
alarmType	Number		Yes	Alarm type. Refer: Alarm of event	
alarmGuid	string	64	Yes	Unique id of alarm message	
alarmGps	string			GPS coordinates when the alarm occurs	113.946564,22.5582962
alarmTime	string		Yes	Alarm occurrence time	2017-01-23 12:12:12
deviceName	String			Device name	
ioName	String			IO name	
alarmValueq	String			Alarm details	
alarmFile	Array			Evidence file	
driverName	String			Driver name	
cardNo	String			Driver card number	
faceImagePath	String			Driver photo	
driverLicenseNo	String			Driver license number	
driverLicensePath	String			Driver license photo	
tel	String			Contact number	
alarmTimeEnd	String			Alarm end time	
speed	Float			Speed	
endGps	String			Alarm end GPS	
takeType	Number			False alarm flag	
reviewType	Number			Cloud re-review result	
fenceId	String			Geofence ID	
fenceName	String			Geofence name	
alarmFile	Object array		Yes	The array of evidence files corresponding to the alarm	
alarmFile Object array:					
fileType	number		Yes	1-Ordinary video, 2-Alarm video, 3-Normal picture, 4-Alarm picture	
filePath	String	256	Yes	Path of evidence file	
channel	number			Channel number, starting from 1	
fileThumbnail	string	256		Evidence thumbnail path	
fileStartTime	string		Yes	Starting time	2017-01-23 12:12:12
fileStopTime	string		Yes	End Time	2017-01-23 12:12:12
downUrl	String	1024		Video download address	
fileGuid	String			Evidence file GUID	

downTaskType	String			File download type	
alarmType	String			Alarm type	
deviceID	String			Device ID	

Request returned data successfully

```
{
  "status": 10000,
  "msg": "success",
  "data": [
    {
      "deviceID": "8888881",
      "alarmGuid": "91BAE324-0B6747A7-BAD0ECED-3EAA4BE0",
      "alarmType": "132",
      "alarmGps": "113.946548,22.5582657",
      "alarmTime": "2021-04-23 13:37:55",
      "alarmFile": [
        {
          "fileGuid": "FB07E391-4FCD44C3-92FEB1BF-475D7188",
          "fileType": "2",
          "filePath": "C:/Program Files
(x86)/HowenVssService/htdocs/vssFiles/hftp/8888881/AI/20210423/133755_82/0_64_82_0_1619185075.mp4",
          "downTaskType": "3",
          "channel": 0,
          "fileThumbnail": "C:/Program Files
(x86)/HowenVssService/htdocs/vssFiles/hftp/8888881/AI/20210423/133755_82/0_64_82_0_1619185075.jpg",
          "fileStartTime": "2021-04-23 13:37:49",
          "fileStopTime": "2021-04-23 13:38:00",
          "alarmType": "132",
          "downUrl":
"https://47.115.83.1:36301/fileSrv/fileDown.php?filePath=QzovUHJvZ3JhbSBGaWxlcyAoeDg2KS9Ib3dlblZzc1NlcnZpY2UvaHRkb
2NzL3Zzc0ZpbGVzL2hmdHAvODg4ODg4MS9BSS8yMDIxMDQyMy8xMzM3NTVfODIvMF82NF84Ml8wXzE2MTkxODUwNz
UubXA0&token=d648caedf89723d96dc114b0dd43e1ea"
        },
        {
          "fileGuid": "1EB9CA53-B175429E-857BD752-288F3F9D",
          "fileType": "4",
          "filePath": "C:/Program Files
(x86)/HowenVssService/htdocs/vssFiles/hftp/8888881/AI/20210423/133755_82/0_64_82_0_1619185075_0.jpg",
          "downTaskType": "3",
          "channel": 0,
          "fileThumbnail": "C:/Program Files
(x86)/HowenVssService/htdocs/vssFiles/hftp/8888881/AI/20210423/133755_82/0_64_82_0_1619185075_0.jpg",
          "fileStartTime": "2021-04-23 13:37:55",
          "fileStopTime": "2021-04-23 13:37:55",
          "alarmType": "132",
          "downUrl":
"https://47.115.83.1:36301/fileSrv/fileDown.php?filePath=QzovUHJvZ3JhbSBGaWxlcyAoeDg2KS9Ib3dlblZzc1NlcnZpY2UvaHRkb
2NzL3Zzc0ZpbGVzL2hmdHAvODg4ODg4MS9BSS8yMDIxMDQyMy8xMzM3NTVfODIvMF82NF84Ml8wXzE2MTkxODUwNz
VfMC5qcGc=&token=de7a6331c39317eb3649654b08cb22"
        }
      ]
    },
    {
      "status": 10003,
      "msg": "Server Error",
      "data": null
    }
  ]
}
```

Other conditions

```
{
  "status": 10003,
  "msg": "Server Error",
  "data": null
}
```

3.13 Device control

Interface address: <http://vss.ptdigital.co.id/vss/vehicle/apiVehicleControl.action>

Request method: http POST

Data format: application/json or application/x-www-form-urlencoded

Private Request parameters: Json

Parameters	Type	Length	Necessary	Details	Example value
deviceID	string	64	yes	Device ID	dev00001
action	number		yes	1--restart 2--Restore factory settings 3--Pan-tilt 4--Image capture 5--Restore the oil circuit 6--cut off the oil circuit 7--Restore circuit 8--cut off the circuit 9--G-Sensor Calibration 10--Send text 12--Reset mileage	
payload[key]	String		No	Instruction parameters, different instructions have different key values. 【Send Text】 Parameters: There are two parameters, namely: (1) payload [type], usually 1 (2) payload [content], is the content of the sent text 【Reset mileage】 parameter Payload [mile], is the reset mileage, in km	
Detailed parameters of image capture command					
channelList	string	64	yes	Channel list, pass; split, start from 1. Only valid for snapshot	1;3;5;6

Request a json format example 1:

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "deviceID": "9999999",
  "action": "4",
  "channelList": "1;4;5;6"
}
```

Request a json format example 2 (Reset mileage):

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "deviceID": "9999999",
  "action": "12",
  "payload[mile]": "100"
}
```

Request a json format example 3 (Send Text):

```
{
  "token": "65292d4efac748beb1f2f530bd13f99b",
  "deviceID": "9999999",
  "action": "10",
  "payload[type]": "1",
  "payload[content]": "Good!"
  "channelList": "1;2"
}
```

Request returned data successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Examples of returned data in other cases

```
{
  "status": 10003,
  "msg": "Server error", //remind information
}
```

```

    "data": null
}

```

3.14 Device control command transparent transmission (supported above VSS V1.7)

This interface is used to transparently transmit device control commands. The commands are encapsulated in json format, and the vss platform forwards the instructions to the device.

Interface address: <http://vss.ptdigital.co.id/vss/vehicle/apiVehicleCommonControl.action>

Request method: http Post

Request data format: json

Private request parameters:

parameter	type	length	Required?	description	Sample value
deviceID	string	64	Yes	device ID	dev00001
action	number		Yes	Command type, currently supports following commands: 1--restart 2--Restore factory settings 3--PTZ 4--Image capture 5--Restore the oil 6--cut off the oil 7--Recovery circuit 8--cut off the circuit 9--G-SENSOR calibration 10- send text	1
Instruction parameter object					
payload	Object		yes	necessary	

Description of each command parameter object

Snapshot command parameters		
Content	Field name	Instruction
Channel list	channelList	Pass; split, starting from 1, such as "1;3;5;6" is 4 channels

"Payload":

```

{
    "channelList": "1;4;5;6"
}

```

PTZ control load parameters		
Content	Section name	instruction
Cction	action	refer to: PTZ action code
Channel	channel	0~11 start from 0
X speed	xspeed	Movement speed in X direction, 1~10
Y direction speed	yspeed	Movement speed in Y direction, 1~10
Preset position	preset	0~255

"Payload":

```

{
    "action": "3",
    "xspeed": "5",
    "yspeed": "5",
    "preset": "1",
    "channel": 0
}

```

Send text load parameters		
Content	Field name	Description
type	type	1- Display text 2- Broadcast text
Text content	content	

Example of request json format:

```
{
  "token": "65292d4efac748beb1f2f530bd
13f99b", "deviceID": "9999999",
  "action": "4",
  "payload": {
    //Command parameter object
    "action": "3",
    "xspeed": "5",
    "yspeed": "5",
    "preset": "1",
    "channel": 0
  }
}
```

Returned data after requesting successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Examples of returned data in other cases

```
{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}
```

3.15 Device video download in Web

3.15.1 Add video download task

The client requests the management center to create a web video download task through this interface.

http interface address: <http://vss.ptdigital.co.id/vss/vss/webdownrecord/insert.action>

Request method: http post

Data format: Content-Type: application/json

Json format parameters of Request

Parameters	Type	Length	Necessary	Details	Example value
token					The same as above
username	String	32	no	Download user	admin
resStartTime	string	64	yes	The start time of the task start time range. Format: YYYY-MM-DD HH:mm:ss	2021-02-23 12:12:11
resEndTime	string		yes	The deadline of the task start time range	
deviceNo	string		yes	Device number, can be empty	
channel	Number		yes	Channel number, starting from 1	
resDeviceFilename	String			The path of the video file on the downloaded device. If this field has a value, it will be downloaded according to the file, otherwise it will be downloaded according to the time range by default	
fileFormat	String		yes	File format, including: ● mp4 ● avi ● csv Currently only supports mp4/csv. Csv is for downloading GPS	
mediaType	String		no	Used when downloading	

				gps, set to gps	
retryCount	Number		no	Maximum retry count on failure	
stream	Number		no	0 - Substream, 1 - Mainstream, default is Mainstream	

Json format example of request:

```
{
  "username": "",
  "deviceNo": "girlfriend",
  "channel": "1",
  "resStartTime": "2021-04-06 00:00:00",
  "resEndTime": "2021-04-06 00:02:00",
  "resDeviceFileName": "",
  "fileFormat": "mp4",
  "token": "f02cfa1cd21138ac89f7f1a32f8339be",
  "lang": "zh_CN"
}
```

Request returned data successfully

```
{
  "status": 10000, //return code, refer to Error code for HTTP interface
  "msg": "success", //return message
  "data": {
    "taskId": "20198002_20201212102810_20201212103810_1_mp4",
    "fileServerHost": "http://172.16.50.210:9988",
    "resWebPath": "/vssFiles/webdown/20198002/20201212102810_20201212103810_1.mp4",
    "repeat": 0
    "taskCreateTime": "2021-04-06 00:02:00"
  }
}
```

Returned data object parameters:

Parameters	Type	Length	Necessary	Details	Example value
taskId	String	32	yes	The task UUID generated by the server.	
fileServerHost	String	128	yes	Video file server address Format :http://ip:port, or: https	
resWebPath	string	256	yes	The web address of the final video file, which compose a complete video download address with fileServerHost	
repeat	Number		yes	0- represents a new task 1- represents a task that is repeated and has been downloaded 2- Represents the repeated task being downloaded Regardless of whether it is repeated, it should be created successfully, it should be displayed in the task list that is being downloaded, and the download progress should be retrieved.	
taskCreateTime	string		yes	Time when the task was created (vss server time)	

Examples of returned data in other cases

```
{
  "status": 10003, //status code, refer to Error code for HTTP interface
  "msg": "Server Error", //remind information
  "data": null
}
```

3.15.2 Video download progress acquisition interface

The client obtains the web video download progress from the management center through this interface.

http interface address: http://vss.ptdigital.co.id/vss/webdownrecord/findTaskProgress.action

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Json format parameters of request

Parameters	Type	Length	Necessary	Details	Example value
token					
taskId	String	128	yes	Task UUID	
resWebPath	string	256	yes	The web address of the final video file.	
token	String		yes	token	

Returned data after requesting successfully

```
{
  "status": 10000, //Return code, refer to Error code for HTTP interface
  "msg": "success", //Return message
  "data": {
    "taskId": "20198002_20201212102810_20201212103810_1_mp4",
    "downState": -1,
    "progress": 81,
  }
}
```

Returned data object parameters:

Parameters	Type	Length	Necessary	Details	Example value
taskId	String	128	yes	Task UUID	
downState	number	128	yes	Download status. 0 not started 1 is downloading 2 Download complete -1 Pause state -7 Device no data -8 Device not online -9 Abnormal stop state When the status is 0/-1/-9, the user can choose to continue downloading	
progress	string	256	yes	The download progress percentage. Integer from 0-100	

-7, -8 is supported by VSS 1.8.5 and later versions.

Examples of returned data in other cases

```
{
  "status": 10003, // status code, refer to Error code for HTTP interface
  "msg": "Server Error", //remind information
  "data": null
}
```

3.15.3 Video download task management (stop/resume/delete) interfaces

The client uses this interface to manage tasks, including stopping tasks, continuing to download (restore), and delete.

http interface address: http://vss.ptdigital.co.id/vss/webdownrecord/taskManagement.action

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Request json format parameters

Parameters	Type	Length	Necessary	Details	Example value
token					
username	string	32	yes	Download user	1
taskId	string	128		Task ID	
action	Number		yes	-1 means delete task, delete task will delete the corresponding video file. 0 means stop the task	

				1 means resume download	
resWebPath	string	256	yes	The web address of the final video file.	

Json format example of request:

```
{
  ...
}
```

Request returned data successfully

```
{
  "status": 10000, //Return code, refer to Error code for HTTP interface
  "msg": "success", /returned messages
  "data": null
}
```

Examples of returned data in other cases

```
{
  "status": 10003, //Return code, refer to Error code for HTTP interface
  "msg": "Server Error", //remind information
  "data": null
}
```

3.15.4 Query video download task list

Interface address: http://192.168.3.15:9000/vss/webdownrecord/findPage.action

Request method: http Post

Data format: application/json or application/x-www-form-urlencoded

Request parameter

Parameters	Type	Length	Necessa ry	Details	Example value
pageNum	int		yes	Page number, query all - 1	2
pageCount	int		yes	Number per page, query all - 1	20
token	varchar		yes	token	
taskState	Int			Task status. -10 - Unknown error -9 - Abnormal stop -8 - Device offline -7 - Device has no data -1 - Paused state 0 - Waiting for device connection 1 - Unfinished download 2 - The download is complete Empty means search out all 1024 - Command issuance completed	
taskStartTime	string	64	yes	The start time of the task start time range. Format: YYYY-MM-DD HH:mm:ss	2021-02-23 12:12:11
taskEndTime	string			Deadline of the task start time range	
deviceNo	string		yes	Device number, could be empty	
username	String	32	yes	Download user	
scheme	String	32		User browser protocol scheme. It can only be http or https	

Request returned data successfully

```
{
```

```

"status": 10000, //return code, refer to Error code for HTTP interface
"msg": "success", //return information
"data": {
    "totalCount": 11, //total number
    "pageCount": -1, //total number per page, -1 means query all=
    "fromCount": 0, //Start
    "toCount": -2, //End of the current page
    "totalNum": -11, //total pages
    "pageNum": -1, //Current page number
    "dataList": [
        {
            "guid": "20198002_20201212102810_20201212103810_1_mp4" //task id,(taskID)
            "deviceNo": "20198002", //device ID
            "channel": 1, //channel number, start from 1
            "createTime": "2020-11-01 12:22:11", //Task creation time
            "resSTime": "2020-11-01 13:22:11", //Recording start time
            "resETime": "2020-11-01 13:42:11", //Recording end time
            "resDeviceFilePath": "/mnt/123/345/3456/67/1221323.avi",
            "resWebPath": "/vssFiles/webdown/20198002/20201212102810_20201212103810_1.mp4",
            "downUrl": "http://172.16.50.210:9988/api/fileDown.php?filePath=RDovc29mdHdhcmUvaHRkb2
            NzL3Zzc0ZpbGVzL3dlYmRvd24vMjAxOTgwMDJfMjAyMTAzMTYwMDAwMDBiMjAyMTAzMTYwMDA4MDBiMS5
            tcDQ=&token=b791530c09905a8ef421e3a1c1b92521",
            "taskState": 1 //1-not finished, 2-finished
        }
    ]
}
}

```

~~downUrl composition:~~

~~1 filePath is a string (filePathBase64) after the base64 of the file path. The path supports full absolute path and web absolute path;~~

~~2 token, should be md5(privatekey+filePathBase64)~~

Examples of returned data in other cases

```

{
    "status": 10003, //status code, refer to Error code for HTTP interface
    "msg": "Server Error", //remind information
    "data": null
}

```

3.16 Passenger flow statistics report

3.16.1 Passenger flow details query

Interface address: http://ip:9966/vss/passengerrecord/detail.action

Request method: http Post

Request data format: json

Private request parameters:

Parameter	Type	length	Required?	Description	Sample value
pageNum	number		Yes	page number	2
pageCount	number		Yes	Number per page	20
devices	string		Yes	Device number, separated by multiple commas	300330,300331
startTime	string		Yes	Starting time	2017-01-23 12:12:12
endTime	string		Yes	End Time	2017-01-23 13:12:12
token	string		Yes	Access token	a559f5ff8af84f08b5e06075193b952f

Example of request json format:

```

{
    "pageNum": "2",
    "pageCount": "20",
    "token": "{{token}}",
    "devices": "99990001,99990002",
    "startTime": "",

```

```
"endTime":""
}
```

Private response parameters:

Returns the json array object files, the parameters of the array element object are as follows:

Parameter	type	length	Required?	Description	Sample value
deviceNo	string	64	Yes	device ID	
fleetName	string		Yes	Fleet name	
driverName	string		no	Driver's name	
createTime	string		no	Creation time	
reportTime	String		Yes	Reporting time	
gps	String		Yes	Location	
dw0	Integer		Yes	Number of people getting off at the front door	
Up0	Integer		Yes	Number of people getting on the front door	
Dw1	Integer		Yes	Number of people getting off the back door	
Up1	Integer		Yes	Number of people getting on the back door	
DwMid	Integer		Yes	Number of people getting off at the middle door	
upMid	Integer		Yes	Number of people getting on the middle door	
curTotal	Integer		Yes	Total number of people in the vehicle	
upTotal	Integer		Yes	Total number of people getting on	
dwTotal	Integer		Yes	Total number of people getting off	
driverCardId	String			Driver card ID	
fleetId	String			Fleet ID	

Request returned data successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": null
}
```

Examples of returned data in other cases

```
{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}
```

3.16.2 Passenger flow statistics query

Interface address: http://localhost:9966/vss/passengerrecord/statistics.action

Request method: http Post

Data format: application/json

Private request parameters:

Parameter	type	length	Required?	Description	Sample value
pageNum	number		Yes	page number	2
pageCount	number		Yes	Number per page	20
devices	string		Yes	Device number, separated by multiple commas	300330,300331
startTime	string		Yes	Starting time	2017-01-23 12:12:12
endTime	string		Yes	End Time	2017-01-23 13:12:12
token	string		Yes	Access token	a559f5ff8af84f08b5e06075193b952f

Example of request json format:

```
{
```



```

"pageNum": "2",
"pageCount": "20",
"token": "{token}",
"devices": "99990001,99990002",
"startTime": "",
"endTime": ""
}

```

Private response parameters:

Return the json array object list, the parameters of the array element object are as follows:

Parameter	type	length	Required?	Description	Sample value
deviceNo	string	64	Yes	device ID	
driverName	string		no	Driver's name	
dw0	Integer		Yes	Total number of people getting off at the front door	
Up0	Integer		Yes	Total number of people getting on at the front door	
Dw1	Integer		Yes	Total number of people getting off at the back door	
Up1	Integer		Yes	Total number of people getting on the back door	
upTotal	Integer		Yes	Total number of people getting on	
dwTotal	Integer		Yes	Total number of people getting off	
fleetId	String			Fleet ID	
fleetName	String			Fleet name	
driverCardId	String			Driver card number	
driverName	String			Driver name	
upMid	Integer			Number of passengers getting on from the middle door	
dwMid	Integer			Number of passengers getting off from the middle door	
reportTime	String			Data reporting time	

Request returned data successfully

```

{
  "status": 10000,
  "msg": "Success",
  "data": null
}

```

Examples of returned data in other cases

```

{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}

```

3.17 Driver report

3.17.1 Query Driver's trip details

Interface address: http://localhost:9966/vss/driver/findDriverTripDetail.action

Request method: http Post

Request data format: json

Private request parameters:

Parameter	type	length	Required?	Description	Sample value
pageNum	number		Yes	page number	2
pageCount	number		Yes	Number per page	20
drivers	string		Yes	Driver card number, separated by multiple commas	test5555,test3333

deviceId	String		No	Device number, must be the complete device number	
startTime	string		Yes	Starting time	2017-01-23 12:12:12
endTime	string		Yes	End Time	2017-01-23 13:12:12
lang	string		Yes	language	zh_CN
token	string		Yes	Access token	a559f5ff8af84f08b5e06075193b952f
minMileage	Float		No	Minimum mileage	

Example of request json format:

```
{
  "pageNum": "2",
  "pageCount": "20",
  "lang": "zh_CN",
  "token": "{{token}}",
  "drivers": "test5555,test3333",
  "startTime": "",
  "endTime": ""
}
```

Private response parameters:

Return the json array object list, the parameters of the array element object are as follows:

parameter	type	length	Required?	Description	Sample value
deviceId	string	64	Yes	device ID	
deviceName	string		Yes	Equipment name	
driverId	string		no	Driver id	
driverName	string		no	Driver's name	
startTime	String		Yes	Start time of the trip	
endTime	String		Yes	End time of the trip	
startGps	String		Yes	Longitude and latitude of the starting point, format: longitude, latitude	
endGps	String		Yes	Longitude and latitude of destination, format: longitude, latitude	
duration	String		Yes	Total travel time, format: HH:mm:ss	
mileage	String		Yes	Total mileage of trip	
speedAvg	String		Yes	Average speed	
speedMax	String		Yes	Maximum speed	
reportTime	String		Yes	Reporting time	
cardNo	String		Yes	card number	
totalAlarmN	Integer			Total number of regular alarms	
totalAlarmDMS	Integer			Total number of DMS alarms	
totalAlarmADAS	Integer			Total number of ADAS alarms	

Request returned data successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "list": [
      {
        "guid": "060D10CB-182D410D-A7E309A6-22573BA2",
        "deviceId": "999",
        "deviceName": "999",
        "driverId": "12232434",
        "driverName": null,
        "startTime": "2021-12-09 18:24:18",
        "endTime": "2021-12-09 18:28:50",
        "startGps": "22.558199,113.946419",

```

```

        "endGps": "113.946701,22.5581665",
        "duration": "00:04:32",
        "timer": null,
        "mileage": 0,
        "speedAvg": "0",
        "speedMax": "0",
        "reportTime": "2021-12-09 18:28:50.0",
        "cardNo": null,
        "totalAlarmN": 3,
        "totalAlarmDMS": 0,
        "totalAlarmADAS": 0
    },
    //....more
},
}
}

```

Examples of returned data in other cases

```

{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}

```

3.17.2 Driver's trip statistics query

interface address: http://localhost:9966/vss/driver/findDriverTripStatistics.action

Request method: http Post

Data format: application/json

Private request parameters:

Parameter	type	length	Required?	Description	Sample value
pageNum	number		Yes	Page number, set to -1 when querying all	2
pageCount	number		Yes	Number per page, set to -1 when querying all	20
drivers	string		Yes	Driver card number, separated by multiple commas	test5555,test3333
deviceId	String		No	Device number, must be the complete device number	
startTime	string		Yes	Starting time	2017-01-23 12:12:12
endTime	string		Yes	End Time	2017-01-23 13:12:12
lang	string		Yes	language	zh_CN
token	string		Yes	Access token	a559f5ff8af84f08b5e06075193b952f

Example of request json format:

```

{
  "pageNum": "2",
  "pageCount": "20",
  "lang": "zh_CN",
  "token": "{{token}}",
  "drivers": "test5555,test3333",
  "startTime": "",
  "endTime": ""
}

```

Private response parameters:

Return the json array object list, the parameters of the array element object are as follows:

parameter	type	length	Required?	Description	Sample value
deviceId	string	64	Yes	device ID	
deviceName	string		Yes	Equipment name	

driverId	string		no	Driver id	
driverName	string		no	Driver's name	
quantity	String		Yes	Number of trips	
duration	number		Yes	Total travel time, in seconds	
mileage	number		Yes	Total mileage of trip	
maxMileage	number		Yes	Maximum mileage	
speedAvg	String		Yes	Average speed	
speedMax	String		Yes	Maximum speed	
reportTime	String		Yes	Reporting time	
cardNo	String		Yes	card number	
totalAlarmN	Integer			Total number of regular alarms	
totalAlarmDMS	Integer			Total number of DMS alarms	
totalAlarmADAS	Integer			Total number of ADAS alarms	
startTime	String			Trip start time	
endTime	String			Trip end time	

Request returned data successfully

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "list": [
      {
        "deviceId": "999",
        "deviceName": "999",
        "driverId": "",
        "driverName": null,
        "quantity": 5,
        "mileage": 67,
        "duration": 1941,
        "maxMileage": 58,
        "speedAvg": "0.00",
        "speedMax": "5",
        "cardNo": null,
        "totalAlarmN": 55,
        "totalAlarmDMS": 0,
        "totalAlarmADAS": 0
      }
    ]
  }
}
```

Examples of returned data in other cases

```
{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}
```

3.18 Geofence (Specifically for T projects)

3.18.1 Add Geofence group

Interface address: <http://localhost:9966/vss/area/addGroup.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
groupName	string		Yes	Group name	Test
lang	string		No	Language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "token": "a559f5ff8af84f08b5e06075193b952f",
  "groupName": "test"
}
```

Response Parameter:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	Group guid, (Unique ID)	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "guid": "16907E37-496043C0-969F4F0B-D29B15B4" //Return group guid
  }
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.2 Delete Geofence group

Interface address: <http://localhost:9966/vss/area/deleteGroup.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
groupGuid	string		Yes	Group ID	16907E37-496043C0-969F4F0B-D29B15B4
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "token": "a559f5ff8af84f08b5e06075193b952f",
  "groupGuid": "16907E37-496043C0-969F4F0B-D29B15B4"
}
```

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success"
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.3 Modify Geofence Group

Interface address: <http://localhost:9966/vss/area/updateGroup.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
-----------	------	--------	-----------	-------------	---------------

groupGuid	string		Yes	Group ID	16907E37-496043C0-969F4F0B-D29B15B4
groupName	string		Yes	Name of new Group	NewTest
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "token": "a559f5ff8af84f08b5e06075193b952f",
  "groupGuid": "16907E37-496043C0-969F4F0B-D29B15B4",
  "groupName": "NewTest"
}
```

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success"
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.4 Get details of Group

Interface address: <http://localhost:9966/vss/area/findGroup.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
groupName	String		No	The name of the searched group, if it is empty, it means to get all the groups	test
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "token": "a559f5ff8af84f08b5e06075193b952f"
}
```

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": [
    {
      "groupGuid": "F6D121D7-37B4406E-A012AA65-FA713461",
      "groupName": "Test"
    }
  ]
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.5 Add Geofence

Interface address: <http://localhost:9966/vss/area/insert.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
areaName	string		Yes	Name of Geofence	
areaType	number		Yes	Geofence Type, 1: polygon 2: route 3: circle	
description	string		No	Geofence Description	
areaInfo	string		Yes	Geofence location data (JSON)	{\"radius\":20848.064,\"gpsData\":[{\"longi\":113.88771,\"lati\":22.78631}]}
areaInfo.radius	number		No	The radius of the circle, if it is a polygon and a path, this field can not be set	
areaInfo.gpsData	array		Yes	latitude and longitude array	
groupGuid	string		No	ID of group	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f
fillColor	String		No	fill color	
lineColor	String		No	border color	

JSON example of request:

```
{
  "areaName": "test",
  "areaType": "1",
  "groupGuid": "E6AAD075-EAC94BE4-BB397971-A9978D43",
  "description": "",
  "areaInfo":
    {\"radius\":0,\"gpsData\":[{\"longi\":113.9050568928826,\"lati\":22.60300073854558},{\"longi\":114.0162934651482,\"lati\":22.64124808536007},{\"longi\":114.02178662921068,\"lati\":22.55198755038766}]},
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b57"
}
```

Response Parameter:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	Geofence guid, (Unique ID)	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data":
    {
      "guid": "16907E37-496043C0-969F4F0B-D29B15B4" //Return Geofence guid
    }
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.6 Get all Geofence (including Group)

Interface address: <http://localhost:9966/vss/area/findAll.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Len	Require	Description	Example Value
-----------	------	-----	---------	-------------	---------------

		gth	d?		
keyword	String		No	Search by Group name or Geofence name. Empty means get all	test
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "token": "a559f5ff8af84f08b5e06075193b952f"
}
```

Response Parameter:

Returns a list of json array objects, and the parameters of the array element objects are as follows:

Parameter	Type	Length	Required?	Description	Example Value
groupGuid	string		Yes	Group ID	
groupName	string		Yes	Group name	
treeList	List		No	Geofence in group	
treeList.areaId	string		Yes	Geofence ID	
treeList.areaName	string		Yes	Geofence name	
treeList.areaType	number		Yes	Geofence Type, 1: Polygon 2: route 3: Circle	
treeList.description	string		No	Geofence Description	
treeList.createUsername	string		Yes	Creator	
treeList.creatoroleid	string		Yes	Role ID of creator	
treeList.groupName	string		No	Group name	
treeList.groupGuid	string		No	Group ID	
fillColor	String		No	fill color	
lineColor	String		No	border color	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": [
    {
      "groupGuid": "F6D121D7-37B4406E-A012AA65-FA713461",
      "groupName": "Test",
      "treeList": [
        {
          "areaId": "04660457-8B834435-A4FC2D52-EA251D1A",
          "areaName": "YandexTest",
          "areaType": 1,
          "description": "",
          "createUsername": "admin",
          "creatoroleid": "3c686465373948d08bc68bfe3b749b20",
          "groupName": "Test",
          "groupGuid": "F6D121D7-37B4406E-A012AA65-FA713461"
        }
      ]
    }
  ]
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.7 Modify Geofence

Interface address: <http://localhost:9966/vss/area/update.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	Geofence ID	
areaName	string		Yes	Geofence name	
areaType	number		Yes	Geofence Type, 1: Polygon 2: route 3: Circle	
description	string		No	Geofence Description	
areaInfo	string		Yes	Geofence location data (JSON)	{\"radius\":20848.064,\"gpsData\": [{\"longi\":113.88771,\"lati\":22.78631}]}
groupGuid	string		No	Group ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f
fillColor	String		No	fill color	
lineColor	String		No	border color	

JSON example of request:

```
{
  "guid": "E6AAD075-EAC94BE4-BB397971-A9978D43",
  "groupGuid": "5D05E347-3B1A-46C9-834E-33B43736B8D4",
  "areaType": "3",
  "areaName": "test2",
  "description": "",
  "areaInfo": "{\"radius\":1516.5209044359387,\"gpsData\": [{\"longi\":113.95087937229425,\"lati\":22.561781465741564}]}",
  "areaId": "E929E9C4-DAB14702-9604F5C4-88329C45",
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b77"
}
```

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success"
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.8 Delete Geofence

Interface address: <http://localhost:9966/vss/area/delete.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guid	String		Yes	Geofence ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "guid": "E6AAD075-EAC94BE4-BB397971-A9978D43",
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b77"
}
```

Return data after request succeeds:

```
{
  "status": 10000,
```

```

    "msg": "Success"
  }
}
Example of return data in other cases:
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}

```

3.18.9 Get details of Geofence

Interface address: http://localhost:9966/vss/area/find.action

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guid	String		Yes	Geofence ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```

{
  "guid": "E6AAD075-EAC94BE4-BB397971-A9978D43",
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b77"
}

```

Response Parameter:

Parameter:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	Geofence ID	
areaName	string		Yes	Geofence name	
areaType	number		Yes	Geofence Type, 1: Polygon 2: route 3: Circle	
description	string		No	Geofence Description	
createUsername	string		Yes	Creator	
creatoroleid	string		Yes	ID of Creator's role	
groupName	string		No	Group name	
groupGuid	string		No	Group ID	
fillColor	String		No	fill color	
lineColor	String		No	border color	

Return data after request succeeds:

```

{
  "status": 10000,
  "msg": "Success",
  "data": {
    "guid": "3C6A7ABA-E48245B4-9B75456E-114233D8",
    "areaname": "GEO FENCE TEST 1",
    "areatype": 1,
    "areainfo": "{\"radius\":0,\"gpsData\":[{\"longi\":113.14970634465196,\"lati\":22.902497017341833}]}",
    "description": "no enter",
    "createtime": "2022-07-14 15:20:01",
    "modifytime": "2022-07-14 15:20:01",
    "creatoroleid": "FC628DCC-C16F43D4-B561BA47-4B1DE48C",
    "groupGuid": "37D0B1DE-7CAF4ABD-92CFA3A1-120D0CFF",
    "createUsername": "demo1"
  }
}

```

Example of return data in other cases:

```

{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}

```

}

3.18.10 Import Geofence from kml

Interface address: http://localhost:9966/vss/area/import.action

Request Method: http Post

Data format: multipart/form-data

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
file	File		Yes	file, supports kml or kmz,xls	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success"
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.11 Batch export to Geofence kml file

Interface address: http://localhost:9966/vss/area/export.action

Request Method: http post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guidList	string		Yes	Geofence ID, multiple IDs are separated by commas	E6AAD075-EAC94BE4-BB397971-A9978D43,CF8E279C-DE4947DD-AC0173CA-3199645F
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f
exportFormat	Integer		No	Export file format, 0: kml 1: xls, default kml format	

JSON example of request:

```
{
  "guidList": "E6AAD075-EAC94BE4-BB397971-A9978D43,CF8E279C-DE4947DD-AC0173CA-3199645F",
  "lang": "zh_CN",
  "token": "18ee7e8cc34d44b38a340b46fc9c9b77"
}
```

Return data after request succeeds:

A kml file is returned, and the Content-Type of the response body is: application/vnd.google-earth.kml+xml or application/octet-stream

The transfer uses block encoding: Transfer-Encoding: chunked

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.12 Issue Geofence to devices

Interface address: http://localhost:9966/vss/areaDevice/insert.action

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
areaId	string		Yes	Geofence ID	
deviceId	string		Yes	Device ID	
toggleConditionJson	string		Yes	Send configuration Parameter (JSON). For configuration Parameter, please refer to the H-protocol document	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "areaId": "3C6A7ABA-E48245B4-9B75456E-114233D8",
  "deviceId": "028073",
  "toggleConditionJson": {
    "over": {
      "enable": 0,
      "spd": "",
      "dut": "",
      "linkRecord": 0,
      "linkLockChn": 0,
      "linkUploadChn": 0,
      "linkOutput": 0,
      "linkSnapChn": 0,
      "linkNet": 0,
      "linkCamSwitch": 0,
      "linkTTSSwitch": 0,
      "linkBuzzer": 0
    },
    "st1": "00:00:00",
    "et1": "23:59:59",
    "st2": "",
    "et2": ""
  },
  "token": "4cb2e1c693254eff87e038db3662213a"
}
```

Response Parameter:

Parameter	Type	Length	Required?	Description	Example Value
guid	string		Yes	guid of Issue record, unique ID of the mapping relationship between Geofence and device	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    {
      "guid": "16907E37-496043C0-969F4F0B-D29B15B4" // Guid of Issue record
    }
  }
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.18.13 Get issue record of a geofence

This interface is used to obtain which devices a Geofence is issued to, and the detailed parameters issued to each device;

Interface address: /vss/areaDevice/getAreaListByAreaId.action

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
areaId	string		Yes	Geofence ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```
{
  "areaId": "3C6A7ABA-E48245B4-9B75456E-114233D8",
  "token": "4cb2e1c693254eff87e038db3662213a"
}
```

Response Parameter:

Returns a list of json array objects, and the parameters of the array element objects are as follows:

Parameter	Type	Length	Required?	Description	Example Value
areaId	string			Geofence ID	
guid	string			guid of Issue record, unique ID of the mapping relationship between Geofence and device	
deviceid	string			Device ID	
createtime	string			Issue time	
modifytime	string			Modify time	
devicename	string			Device name	
areaSendState	number			Status: 0: not synched 1: synched	
areaSendErrCode	string			error code, 0: not issued or issued successfully 1: issuing timed out -1: json generation failed 99: other errors	
toggleconditionjson	string			configuration Parameter of issuing (JSON). For configuration Parameter, please refer to the H-protocol document	
lastToggleConditionJson	string			Last configuration Parameter of issuing (JSON). For configuration Parameter, please refer to the H-protocol document	

Return data after request succeeds:

```
{
  "status": 10000,
  "msg": "Success",
  "data": [
    {
      "guid": "74A5CD7F-C779460D-8ABF3CB0-85BE2DD2",
      "areaid": "3C6A7ABA-E48245B4-9B75456E-114233D8",
      "deviceid": "028073",
      "toggleconditionjson":
      "{\r\n  \"over\": {\r\n    \"enable\": 0,\r\n    \"spd\": \"\", \r\n    \"dut\": \"\", \r\n    \"linkRecord\": 0, \r\n    \"linkLockChn\": 0, \r\n    \"linkUploadChn\": 0, \r\n    \"linkOutput\": 0, \r\n    \"linkSnapChn\": 0, \r\n    \"linkNet\": 0, \r\n    \"linkCamSwitch\": 0, \r\n    \"linkTTSSwitch\": 0, \r\n    \"linkBuzzer\": 0\r\n  },\r\n  \"createtime\": \"2022-07-14 17:02:39\", \r\n  \"modifytime\": \"2022-07-14 17:02:39\", \r\n  \"devicename\": \"028073\", \r\n  \"areaSendState\": 1,\r\n  \"areaSendErrCode\": \"\", \r\n  \"toggleconditionjson\": \"\", \r\n  \"lastToggleConditionJson\": \"\"\r\n}"
    }
  ]
}
```

```

        "areaSendErrCode": 0,
        "lastToggleConditionJson":
        "{\over\": {\enable\":0,\"spd\": \"\", \"dut\": \"\", \"linkRecord\":0, \"linkLockChn\":0, \"linkUploadChn\":0, \"linkOutput\":0, \"linkSnapC
hn\":0, \"linkNet\":0, \"linkCamSwitch\":0, \"linkTTSSwitch\":0, \"linkBuzzer\":0}"
    }
]
}

```

Example of return data in other cases:

```

{
    "status": 10003,
    "msg": "Server Error", //Notification info
    "data": null
}

```

3.18.14 Delete a Geofence in device

Interface address: <http://localhost:9966/vss/areaDevice/delete.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Require d?	Description	Example Value
guid	string		Yes	ID of issue record. Note: This is not the Geofence id, but the guid returned by the Geofence interface	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```

{
    "guid": "3C6A7ABA-E48245B4-9B75456E-114233D8",
    "token": "4cb2e1c693254eff87e038db3662213a"
}

```

Return data after request succeeds:

```

{
    "status": 10000,
    "msg": "Success"
}

```

Example of return data in other cases:

```

{
    "status": 10003,
    "msg": "Server Error", //Notification info
    "data": null
}

```

3.18.15 Get the issue details of a Geofence

This interface is used to obtain the detailed parameters of issuing a Geofence to a device.

Interface address: <http://localhost:9966/vss/areaDevice/getlastToggleConditionJson.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
areaId	string		Yes	Geofence ID	
deviceId	string		Yes	Device ID	
lang	string		No	language	zh_CN
token	string		Yes	token	a559f5ff8af84f08b5e06075193b952f

JSON example of request:

```

{
    "areaId": "3C6A7ABA-E48245B4-9B75456E-114233D8",
    "deviceId": "20160604",
    "token": "4cb2e1c693254eff87e038db3662213a"
}

```

```
}
For the specific definition of the returned data, please refer to the H-protocol document.
Return data after request succeeds:
```

```
{
  "status": 10000,
  "msg": "Success",
  "data": {
    "over": {
      "linkSnapChn": 0,
      "dut": "",
      "linkOutput": 0,
      "linkCamSwitch": 0,
      "enable": "1",
      "spd": "",
      "linkUploadChn": 1057,
      "linkBuzzer": 0,
      "linkTTSSwitch": 0,
      "linkLockChn": 1057,
      "linkNet": 0,
      "linkRecord": 0
    }
  },
  ...
}
```

Example of return data in other cases:

```
{
  "status": 10003,
  "msg": "Server Error", //Notification info
  "data": null
}
```

3.19 Remote Configuration of Device Parameters

This interface can only be used when the device is online.

3.19.1 Obtain Device Parameter

Interface address: <http://serverIp:9966/vss/vehicle/getVehicleSetting.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
settingPara	Object		yes	Obtain parameter type	Refer to the "Device Parameter Description Document"
deviceID	string		yes	Device ID	
lang	string		yes	Language	zh_CN
token	string		yes	token	6aae336cd15643b7b24ddcd958cec703

JSON example of request:

```
{
  "deviceID": "dashV8001",
  "settingPara": {
    "VERSIONINFO": "",
    ... // Multiple parameter segment names can be used
  },
  "token": "6aae336cd15643b7b24ddcd958cec703",
  "lang": "zh_CN"
}
```

To define the content of settingPara data, please refer to "Device Communication Protocol_Device Parameter Description Document". Note that settingPara is an Object, which can contain one or more parameter segments. When the value corresponding to the parameter segment is an empty string, it means that a query for all parameters of this parameter section is required..

This example is used to query device version information. The parameter section name of the version information is VERSIONINFO. Return data after request succeeds:

```
{
  "status": 10000,
```

```

"msg": "succeed",
"error": null,
"data": {
  "VERSIONINFO": {
    "alg": "T23092101.1524",
    "app": "T23110101.1550",
    "boot": "V21111001",
    "ext": "T23110101.1550",
    "hardware": "81",
    "kernel": "V230831000",
    "mcu": "G23102710",
    "rootfs": "V23062801"
  }
},
"count": 0
}

```

The returned data object also contains detailed parameter values from 1 to multiple parameter segments. For details, please refer to "Device Communication Protocol_Device Parameter Description Document"

3.19.2 Set device parameters

Interface address: <http://serverIp:9966/vss/vehicle/apiVehicleSetting.action>

Request Method: http Post

Data format: json

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
settingPara	Object		yes	Obtain parameter type	Refer to the "Device Parameter Description Document"
deviceID	string		yes	Device ID	
lang	string		no	Language	zh_CN
token	string		yes	token	6aae336cd15643b7b24ddcd958cec703

JSON example of request:

```

{
  "deviceID": "dashV8001",
  "settingPara": {
    "JTBASE": {
      "DevId": "",
      "ImeiLen": "8",
      "SetDeviceID": "1",
      "Sn0104": "3B4398001812944F",
      "TerminalId": "6646",
      "city": "sz",
      "color": "0",
      "gpsInterval": "0",
      "gpsPosMode": "4",
      "gpsinterval1": "30",
      "gpsinterval2": "30",
      "license": "356TH",
      "manufacturer": "Howen",
      "model": "dashcamV8",
      "phonenum": "dashV8001",
      "protocol1": "0",
      "protocol2": "2",
      "protocol3": "0",
      "protocol4": "0",
      "province": "gd",
      "ttsControl": "0",
      "vehicle_dvr": "",
      "vehicle_motor": "",
      "vehilce_type": "0"
    }
  },
  "token": "6aae336cd15643b7b24ddcd958cec703",
  "lang": "zh_CN"
}

```


}
 To define the content of settingPara data, please refer to "Device Communication Protocol_Device Parameter Description Document".
 Note that settingPara is an Object, which can contain one or more parameter segments. Each parameter segment contains the setting value of each parameter.

For example, this example is used to set basic configuration parameters. The parameter section name of the version information is JTBASE.

Return data after request succeeds:

```

{
  "status": 10000,
  "msg": "succeed",
  "data": {
    "JTBASE": {
      "DevId": "",
      "ImeiLen": "8",
      "SetDeviceID": "1",
      "Sn0104": "3B4398001812944F",
      "TerminalId": "6646",
      "city": "sz",
      "color": "0",
      "gpsInterval": "0",
      "gpsPosMode": "4",
      "gpsinterval1": "30",
      "gpsinterval2": "30",
      "license": "356TH",
      "manufacturer": "Howen",
      "model": "dashcamV8",
      "phonenum": "dashV8001",
      "protocol1": "0",
      "protocol2": "2",
      "protocol3": "0",
      "protocol4": "0",
      "province": "gd",
      "ttsControl": "0",
      "vehicle_dvr": "",
      "vehicle_motor": "",
      "vehilce_type": "0"
    }
  }
}

```

The returned data object contains the parameter value of the corresponding parameter segment after successful setting.

3.20 Driver management

The driver management interface not only creates and manages the driver information within the platform, but also synchronizes the driver information automatically by the VSS platform to the corresponding assigned vehicle equipment. When a driver is assigned to a vehicle, if the driver is deleted or key information such as the card number is modified, the VSS platform will automatically re-synchronize this driver information to the vehicle it is assigned to. If the vehicle is not online, the platform will automatically synchronize the driver information to the device after the vehicle is online next time.

3.20.1 Add new driver

Interface address: <http://serverIp:9966/vss/driver/insert.action>

Request mode: http Post

Data format: multipart/form-data

Private request parameters:

Parameters	Type	Length	Required/ Not Required	Description	Example Value
cardNo	string	64	Required	Driver Car Number	
driverName	string	64	Required	Driver name	
fleetId	string	255	Required	Fleet guid	System default fleet can be used guid 38917B17-9AAA45DB-833FC480-0AA08C42
tel	String		Required	Phone number	
token	String		Required	Token	
sex	String		Required	Sex: 1 Male, 2 Female	
birthDate	String		Not Required	Date of Birth	2004-01-01
driverFaceImage	File		Not Required	Driver Face Image	The format can only be jpg/png, and the file size cannot exceed 1.6M
driverLicenceImage	File		Not Required	Driver License Image	The format can only be jpg/png, and the file size cannot exceed 0.8M
email	String		Not Required	email	
identityId	String		Not Required	ID card number	
licenceNo	String		Not Required	Driver License Number	
remarks	String		Not Required	Remarks	
Version	String		Not Required	VSS version	Note: This is a header field

The request is successful and the data is returned:

```
{
  "status":10000,
  "msg":" success ",
  "error":null,
  "data":null,
  "count":0
}
```

Other cases return data examples:

```
{
  "status": 10003,
  "msg": "server error", //prompt message
  "data": null
}
```

3.20.2 Query driver list

Interface address: <http://serverIp:9966/vss/driver/findGroup.action>

Request mode: http Post

Data format: application/json

Private request parameters:

Parameters	Type	Length	Required/ Not Required	Description	Example Value
driverName	String	64	Not Required	Driver Name	
token	String		Required	Token Certificate	
scheme	String		Not Required	Protocol: http, https	
fleetId	String		Not Required	Fleet ID	
driverId	String		Not Required	driver id	
deviceId	String		Not Required	device id	
fleetName	String		Not Required	Fleet name	

Json Request format example:

```
{
  "driverName": "",
  "token": "f77135a6053e459da3eb5a98844c8af7",
  "scheme": "http"
}
```

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guid	String			Fleet GUID	
fleetname	String			Fleet name	
pid	String			Parent fleet	
contacts	String			Contact person	
tel	String			Phone	
email	String			Email	
createtime	String			Creation time	
modifytime	String			Modification time	
parentfleetname	String			Parent fleet name	
vehiclecount	String			Number of devices	
treeList	Array			Driver list	
Driver info					
id	String			Driver ID	
drivername	String			Driver name	
tel	String			Phone	
sex	Integer			Gender	
age	String			Age	
email	String			Email	
cardno	String			Driver card number	
pid	String			Parent ID	
remarks	String			Remarks	
identityid	String			Identity ID	
faceimagepath	String			Face photo URL	
driverlicenseno	String			Driver license number	
driverlicensepath	String			Driver license photo URL	
birthdate	String			Date of birth	
createtime	String			Creation time	
modifytime	String			Modification time	
fleetname	String			Associated fleet name	
sendState	String			Issuance status (0 - Waiting to issue, 1 - Issue failed, device unresponsive, 2 - Issue failed, information generation failed, 99 - Issue failed, unexpected)	

				error, -1 - Issued)	
--	--	--	--	---------------------	--

The request is successful and the driver information will be returned according to the team group.

```
{
  "status":10000,
  "msg":"succeed",
  "error":null,
  "data":[
    {
      "fleetname":"fleettest",
      "treeList":[
        {
          "id":"084F3527-1B0746A3-A5BC0FA7-8BF9F929", //Driver guid
          "pid": "38917B17-9AAA45DB-833FC480-0AA08C42" //fleet id
          "drivename":"charlie002",
          "tel":"16546595",
          "sex":"1", //Sex: 1Male、2Female
          "age":0,
          "email":"",
          "cardno":"5488", //Driver Card number
          "remarks":null,
          "identityid":"",
        }
      ]
    }
  ],
  "count":0
}
```

Other cases return data examples

```
{
  "status": 10003,
  "msg": "server error", //prompt message
  "data": null
}
```

3.20.3 Remove driver

Interface address: <http://serverIp:9966/vss/driver/deleteBatch.action>

Request mode: http Post

Data format: application/x-www-form-urlencoded

Private request parameters:

Parameters	Type	Length	Required/ Not Required	Description	Example Value
guids	string		Required	Driver guid field. Multiple GUIDs are separated by commas (,)	
token	String		Not Required	Token Certificate	

The request is successful and the data is returned:

```
{
  "status":10000,
  "msg":" success ",
  "error":null,
  "data":null,
  "count":0
}
```

Other cases return data examples:

```
{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}
```

3.20.4 Modify driver

Interface address: http://serverIp:9966/vss/driver/update.action

Request mode: http Post

Data format: multipart/form-data

Private request parameters:

Parameters	Type	Length	Required/ Not Required	Description	Example Value
guid	string	64	Required	Driver guid	
cardNo	string	64	Required	Driver Card Number	
driverName	string	64	Required	Driver Name	
fleetId	string	255	Required	Fleet ID	The system default fleet guid can be used 38917B17-9AAA45DB-833FC480-0AA08C42
tel	String		Required	Phone Number	
token	String		Required	Token Certification	
birthDate	String		Not Required	Date of Birth	2024-01-01
driverFaceImage	File		Not Required	Driver Face Image	If you need to change the format, see Added
driverLicenceImage	File		Not Required	Driver ID picture	If you need to change the format, see Added
email	String		Not Required	email	
identityId	String		Not Required	ID number	
licenceNo	String		Not Required	driver's license number	
remarks	String		Not Required	remarks	
sex	String		Not Required	Gender: 1Male、2Male	
Version	String		Not Required	VSS version	Note: This is a header field

The request is successful and the data is returned

```
{
  "status":10000,
  "msg":" success ",
  "error":null,
  "data":null,
  "count":0
}
```

Other cases return data examples

```
{
  "status": 10003,
  "msg": "Server Error", //Prompt
  "data": null
}
```

3.20.5 Assign drivers to vehicles (bind)

The interface address: http://serverIp:9966/vss/driver/issueDriverInfo.action

Request mode: http Post

Data format: application/json

Private request parameters:

Parameters	Type	Length	Required/ Not Required	Description	Example Value
token	String		Required	Token Certificate	
devices	array		Required	Vehicle terminal device ID, can be passed multiple	["deviceId01", "deviceId02"]

drivers	array		Required	Driver guid, multipassable	[“guid01”, “guid02”]
scheme	String		Required	Internet Protocol: http, https	

Json Request format example:

```
{
  "devices": [
    " deviceId01",
    " deviceId01"
  ],
  "drivers": [
    "6445A4D0-5D4443F4-8777E866-9DAB5744",
    "7E0FCC42-82F14BA2-BB704C09-F811CC58"
  ],
  "token": "f77135a6053e459da3eb5a98844c8af7",
  "scheme": "http",
  "lang": "zh_CN"
}
```

The request is successful and the data is returned

```
{
  "status": 10000,
  "msg": " success ",
  "error": null,
  "data": null,
  "count": 0
}
```

Other cases return data examples

```
{
  "status": 10003,
  "msg": "Server error", //Prompt message
  "data": null
}
```

3.20.6 Remove driver from vehicle (unbind)

The interface address: http://serverIp:9966/vss/driver/deleteDriverInfoByFleet.action

Request mode: http Post

Data format: application/json

Private request parameters:

Parameters	Type	Length	Required/ Not Required	Description	Example Value
token	String		Required	Token Certificate	
deviceId	String		Required	Device ID	
driverList	array		Required	Driver guid, multipassable	[“guid01”, “guid02”]
scheme	String		Required	Network protocol: http, https	

Json Request format example:

```
{
  "deviceId": " deviceId 01",
  "driverList": [
    "6445A4D0-5D4443F4-8777E866-9DAB5744",
    "7E0FCC42-82F14BA2-BB704C09-F811CC58"
  ],
  "token": "f77135a6053e459da3eb5a98844c8af7",
  "scheme": "http",
  "lang": "zh_CN"
}
```

The request is successful and the data is returned

```
{
  "status": 10000,
  "msg": " success ",
  "error": null,
}
```

```

    "data":null,
    "count":0
}
Other cases return data examples
{
    "status": 10003,
    "msg": "Server errors", //Prompt Message
    "data": null
}

```

3.20.7 View the driver list and delivery status of a vehicle:

Interface address: <http://vss.ptdigital.co.id/vss/driver/findDriverInfoByFleet.action>

Request mode: http Post

Data format: application/json

Private request parameters:

Parameters	Type	Length	Required/ Not Required	Description	Example Value
deviceId	String		Required	Device ID	
token	String		Required	Token Certificate	
scheme	String		Not Required	Network Protocol: http, https	

Json Request format example:

```

{
    "deviceId":"666666",
    "token":"f77135a6053e459da3eb5a98844c8af7",
    "scheme":"http"
}

```

Request parameters:

Parameter	Type	Length	Required?	Description	Example Value
guid	String			Fleet GUID	
fleetname	String			Fleet name	
pid	String			Parent fleet	
contacts	String			Contact person	
tel	String			Phone	
email	String			Email	
createtime	String			Creation time	
modifytime	String			Modification time	
parentfleetname	String			Parent fleet name	
vehiclecount	String			Number of devices	
treeList	Array			Driver list	
Driver info					
id	String			Driver ID	
drivername	String			Driver name	
tel	String			Phone	
sex	Integer			Gender	
age	String			Age	
email	String			Email	
cardno	String			Driver card number	
pid	String			Parent ID	
remarks	String			Remarks	
identityid	String			Identity ID	
faceimagepath	String			Face photo URL	
driverlicenseno	String			Driver license number	
driverlicensepath	String			Driver license photo URL	
birthdate	String			Date of birth	

createtime	String			Creation time	
modifytime	String			Modification time	
fleetname	String			Associated fleet name	
sendState	String			Issuance status (0 - Waiting to issue, 1 - Issue failed, device unresponsive, 2 - Issue failed, information generation failed, 99 - Issue failed, unexpected error, -1 - Issued)	

Request success Return data (grouped by fleet)

```
{
  "status":10000,
  "msg":"Success",
  "error":null,
  "data":[
    {
      "fleetname":"fleetttest", //Fleet name
      "treeList":[
        {
          "id":"084F3527-1B0746A3-A5BC0FA7-8BF9F929", //Driver guid
          "drivename":"charlie002",
          "tel":"16546595",
          "sex":"1", //Gender: 1 Male、2 Female
          "age":0,
          "email":"",
          "cardno":"5488", //Driver card number
          "pid": "38917B17-9AAA45DB-833FC480-0AA08C42", //Fleet ID
          "remarks":null,
          "identityid":"",
          "sendState":-1 //Delivery status: 1: "waiting to be issued," 0: "success", 1: "has been successfully issued", 2: "is being downloaded photos", 3: "the download is complete," 4: "photos in check," 10: "failure pictures (illegal)", 11: "failure" (photo download failure), 12: "Failed identification (duplicate photo)",13:" Exceeded face count ", 99: "Unknown"
        }
      ]
    }
  ],
  "count":0
}
Other cases return data examples
{
  "status": 10003,
  "msg": "Server Error", //Prompt Message
  "data": null
}
```

3.20.8 Query Driver Details

Interface address: <http://serverIp:9966/vss/driver/find.action>

Request mode: http Post

Data format: x-www-form-urlencoded

Private request parameters:

Parameters	Type	Length	Required/Not Required	Description	Example Value
guid	String	64	Required	Driver guid	
token	String		Required	Token Certificate	
cardNo	String		Required	Card number	

Request parameters:

Parameters	Type	Length	Required/ Not Required	Description	Example Value
guid	String			Driver GUID	
drivername	String			Driver name	
tel	String			Phone	
age	String			Age	
email	String			Email	
cardno	String			Driver card number	
fleetid	String			Associated fleet ID	
remarks	String			Remarks	
identityid	String			Identity ID	
faceimagepath	String			Face photo URL	
driverlicenseno	String			Driver license number	
driverlicensepath	String			License photo URL	
birthdate	String			Date of birth	
createtime	String			Creation time	
modifytime	String			Modification time	
fleetname	String			Fleet name	
faceMD5	String			Face MD5	
roleNumber	String			Role ID	

The request is successful and the driver information will be returned according to the team group.

```
{
  "status":10000,
  "msg":"Success",
  "error":null,
  "data":{
    //Driver guid
    guid:"15D42C16-B83B4F26-B1E7D33B-6B251DB7"
    drivername "abc"
    tel "01255"
    sex "1"
    age 0
    email ""
    cardno "401"
    fleetid "8C286873-A16046F4-BD9BBD6F-3DAE8BF4"
    remarks null
    identityid ""
    //Driver face address (example: http://serverIp:9966/vss/driver/showImage.action?guid=guid&type=1 )
    faceimagepath:"/driver/showImage.action?guid=15D42C16-B83B4F26-B1E7D33B-6B251DB7&type=1"
    driverlicenseno:""
    //Driver ID photo address (example: http://serverIp:9966/vss/driver/showImage.action?guid=guid&type=2 )
    driverlicensepath:"/driver/showImage.action?guid=15D42C16-B83B4F26-B1E7D33B-6B251DB7&type=2"
    birthdate null
    createtime "2024-01-07 14:38:16"
    modifytime "2024-01-07 14:38:16"
    fleetname "01_TSD_Cody-Demo"
    faceMD5 null
    roleNumber "rv77"
  }
}
Other cases return data examples
{
  "status": 10003,
  "msg": "Server Error", //Prompt message
  "data": null
}
```

3.21 Voice file management

3.21.1 Add voice file

Interface address: http://serverIP:9966/vss /voice/ addVoiceFile.action

Request method: http Post

Data format: multipart/form-data

Private request parameters:

Parameter	Type	Length	Required?	Description	Example value
token	String		yes	Token	
lang	String		no	language type	
voiceFile	File		yes	voice file	A compressed voice file ending in .tar
issueCallbackUrl	String		yes	Voice packet delivery result return address url	http://serverIP:9966/vss/voice/addVoiceFile.action
voicePackageName	String		yes	Voice package name (customized)	
voicePackageNo	String		yes	Voice packet number (unique)	

Data returned after requesting successfully:

```
{
  "status":10000 ,
  "msg":"success",
  "error":null ,
  "data":null ,
  "count":0
}
```

Examples of data returned in other situations

```
{
  "status": 10003,
  "msg": "Server error", // Prompt message
  "data": null
}
```

3.21.2 Delete voice files

Interface address: http://serverIP:9966/vss/voice/deleteVoiceFile.action

Request method: http Post

Data format: application/json

Private request parameters:

Parameter	Type	Length	Required?	Description	Example value
voicePackageNo	string		yes	Voice file unique number	
token	String		yes	Token certification	

Data returned after requesting successfully:

```
{
  "status":10000 ,
  "msg":" success ",
  "error":null ,
  "data":null ,
  "count":0
}
```

Examples of data returned in other situations

```
{
  "status": 10003,
  "msg": "Server error", // Prompt message
  "data": null
}
```

3.21.3 Modify the voice file information (modify according to the voice package number)

Interface address: http://serverIP:9966/vss/voice/updateVoiceFile.action

Request method: http Post

Data format: application/json

Private request parameters:

Parameter	Type	Length	Required?	Description	Example value
token	String		yes	Token	
issueCallbackUrl	String		no	Voice packet delivery result return address url	http://serverIP:9966/vss/voice/addVoiceFile.action
voicePackageName	String		no	Voice package name	

				(customized)	
voicePackageNo	String		yes	Voice packet number (unique)	

Data returned after requesting successfully:

```
{
  "status":10000,
  "msg":" success ",
  "error":null,
  "data":null,
  "count":0
}
```

Examples of data returned in other situations

```
{
  "status": 10003,
  "msg": "Server error", // Prompt message
  "data": null
}
```

3.21.4 Query voice files

Interface address: http://serverIP:9966/vss/voice/ getVoiceFile.action

Request method: http Post

Data format: application/json

Private request parameters:

Parameter	Type	Length	Required?	Description	Example value
token	String		yes	Token certification	

Request parameters:

Parameters	Type	Length	Required/ Not Required	Description	Example Value
voicePackageNo	String		Required	Voice file unique ID	
voicePackageName	String			Voice package name	
voiceFileName	String			Voice file name	
voicePackageSize	String			Voice package size (in bytes)	
createTime	String			Voice file creation time	
updateTime	String			Voice file modification time	
issueCallbackUrl	String			Voice package issuance result callback URL	
createUsername	String			Uploaded username	
roleNumber	String			Role ID	

Data returned after requesting successfully:

```
{
  "status":10000,
  "msg":" success ",
  "error":null,
  "data": [
    {
      "voicePackageName ": "", voice package name (user-defined)
      " createUsername ": "", upload username
      " createTime ": "", upload time
      " voiceFileName ": "", voice file name
      " voicePackageNo ": "", voice package number
      " issueCallbackUrl ": "", voice package delivery result return address url
      " updateTime ": "", last modified time
      " voicePackageSize ": 0, voice file size in bytes
      " roleNumber ": "" The role number of the upload user
    }
  ], " count":0
}
```

Examples of data returned in other situations

```
{
  "status": 10003,
  "msg": "Server error", // Prompt message
  "data": null
}
```

3.21.5 Send voice files to a device

Interface address: http://serverIP:9966/vss /voice/ issueVoiceFile.action

Request method: http Post

Data format: application/json

Private request parameters:

Parameter	Type	Length	Required?	Description	Example value
voicePackageNo	string		yes	Voice file unique number	
deviceId	string		yes	device ID	
sessionId	string		yes	Session ID, the unique number of this session	
token	string		yes	Token certification	

Data returned after requesting successfully:

```
{
  "status":10000 ,
  "msg": " success ",
  "error":null ,
  "data":null ,
  "count":0
}
```

Examples of data returned in other situations

```
{
  "status": 10003,
  "msg": "Server error", // Prompt message
  "data": null
}
```

3.21.6 After the delivery is successful, the VSS platform returns the message definition to FMS

When the VSS is successfully delivered, the issueCallbackUrl callback interface will be called. The following is the request json message body:

```
{
  "sessionId ":" ", //This session number
  "deviceNo ":null, //device number
  "status": 1 , //1 download started, 0 download completed
  "errCode":0 //0 success, non-0 failure
}
```

3.21.7 Issue voice broadcast

Interface address: http://serverIP:9966/vss /voice/ openVoice.action

Request method: http Post

Data format: application/json

Private request parameters:

Parameter	Type	Length	Required?	Description	Example value
pcmNo	string		yes	Sequency No. of pcm voice file to be broadcast	TTS_001.pcm corresponds to 1, and so on.
deviceId	string		yes	device ID	
token	string		yes	Token certification	

Data returned after requesting successfully:

```
{
  "status":10000 ,
  "msg": " success ",
  "error":null ,
  "data":null ,
  "count":0
}
```

Examples of data returned in other situations

```
{
  "status": 1000 7 ,
  "msg": "Failed; errCode=9",
  //errorCode 0: Success 1: Invalid parameter 2: Invalid command 3: Invalid service 4: Service not started 5: Object does not exist
  6: Waiting timeout 7: Connection failed 8: File opening failed 9: The device is not online
  // 10: Device busy 11: Network exception 12: Failed to obtain disk media block 13: Failed to read media block 14: Failed to
  open media file 15: Data decryption failed 254: Not supported 255: Unknown error
}
```

3.22 Traffic records

3.22.1 Traffic consumption record

Interface address: http://serverIP:9966/vss / flowrecord / findPage.action

Request method: http Post

Data format: application/json

Request parameters:

parameter	type	length	Required?	Description	Example value
flowType	Number		no	Traffic type, 1- Signaling 2-Real-time video 3-Remote playback 4- File transmission 5-Alarm linkage video download 6-Scheduled video download	
deviceID	string		no	Device ID, multiple device IDs separated by commas	
direct	Number		no	Traffic direction: 1-device upstream 2-device downstream	
network	Number		no	Network type, 3: 2G, 4: 3G, 5: 4G, 6: 5G	
pageNum	Number		yes	Pagination page number	1
pageSize	Number		yes	Paging Size	50
token	string		yes	Token certification	
startFlowValue	Number		x	start flow value	
endFlowValue	Number		no	end flow value	

Data returned after requesting successfully:

```
{
  "status": 10000,
  "msg": "success",
  "error": null,
  "data": {
    "total": 12,
    "list": [
      {
        "guid": " A4D1399B - F0954F34 - AEE653CF -00A1BBEC",
        "sim": "",
        "deviceNo ": "000057",
        "direct": 1,
        "flowType": 1,
        "consumeTime ": "2024-04-22 11:45:16",
        "network": 5,
        "reportTime ": "2024-04-22 11:45:16",
        "flowValue ": 2288,
        "deviceName": "000057",
        "flowValueText ": 0.0
      }
    ]
  }
}
```

4 Websocket real time information Interface

Before starting WebSocket subscription for real-time business messages, it must also log in through the HTTP login interface, which requires to obtain the PID parameter and token parameter returned by the server from the login interface and use them for WebSocket subscription requests.

Subscribe to real-time information including device online or offline, alerts/events, GPS status, and more.

4.1 ws login

Interface Address: ws://192.168.1.2:36300/ws

Request way: websocket, short as 'ws'

Request direction: ws client->ws server

Private request parameter:

Json Object/Attribute	Type	Length	Required	Description	Example value
action	number		Yes	ws message type, Message type, login and subscription type is 80,000, see: ws message type table	80000
Payload object:					
payload.username	string	64	Yes	User name	admin
payload.pid	string	128	Yes		b9c7f86sdeg6f7e4799cefgaaed465dc1cc704cb1db207bda b9c1758ad39bd187d350fa949b2ff9ddd a8436467b1
payload.token	string	64	Yes	The token obtained after a successful API login	0db6acgf92d0a2bds 5702d92ef906631

Json format example:

```
{
  "action": "80000",
  "payload": {
    "username": "admin",
    "pid": "b9c7f86f5a21dc6f7e4799cde801eed465dc1cc704cb1db207bdab9c1758ad39bd187d350fa949b2ff9ddd8436467b1",
    "token": "0db6d4b92d0a2b0c45702d92ef906631"
  }
}
```

Request successfully and return data

```
{
  "action": "80000",
  "payload": {
    "result": "success",
    "msg": "success"
  }
}
```

Return data example in the other situation

```
{
  "action": "80000",
  "payload": {
    "result": "fail",
    "msg": "Invalid username"
  }
}
```

4.2 ws subscribe to real-time message

interface address: ws://192.168.1.2:36300/ws

Request method: websocket, also named as ws

Request direction: ws client->ws server

Private request parameters:

Json Object	Type	Len	Required	Description	Example
action	number		Yes	ws message type, subscription type is 80001, refer: ws message type table	80001
Payload Object: Empty					

Json format example:

```
{
  "action": "80001",
  "payload": ""
}
```

4.3 Heartbreak message

Interface address: ws://192.168.1.2:36300/ws

Request direction: ws client->ws server

Client should send the heartbreak message to server per 1 minute to make sure channel active.

Private request parameter:

Json Object/Attribute	Type	Length	Required	Description	Example Value
action	number		Yes	ws message type, login and subscription type is 80009, see: ws message type table	
Payload Object:					
payload.username	string	64	Yes	User name	admin
payload.token	string	64	Yes	The token obtained after a successful API login	0db6acgf92d0a2bds5702d92ef906631

Json format example:

```
{
  "action": "80009",
  "payload": {
    "username": "admin",
    "token": "0db6d4b92d0a2b0c45702d92ef906631"
  }
}
```

Request successfully and return data

```
{
  "action": "80009",
  "payload": {
    "result": "success",
    "msg": ""
  }
}
```

4.4 Push device status information

Request way : websocket

Request direction: ws server->ws client

Request parameter:

Json Object/Attribute	Type	length	Required	Description	Example value
action	number		Yes	ws message type, push status message Yes80003, see: WS message type table	80000
Payload object definition					
payload.deviceID attribute	string	128	Yes	Device id	99990001
payload.location positioning information object					
location.mode	Number	8	Yes	Location type 0: invalid positioning 1: GPS 2-BD 3: GLONASS	1

				4: AGPS 5: base station location 6: WIFI	
location.dtu	String	24	Yes	GPS positioning time	2018-01-11 11:35:02
location.direct	Number	8	Yes	Vehicle direction, 0~359°	
location.speed	Number		Yes	Speed, unit: KM/H	
location.altitude	String		Yes	Altitude(meter)	
location.precision	Number		Yes	Positioning precision(meter)	
location.longitude	String		Yes	longitude	114.106476
location.latitude	String		Yes	latitude	22.5767326
payload.module Module state object					
module.mobile	Number			Mobile Network module status, (0:the unknown,1:work,2:abnormal,3: not exist)	
module.location	Number			Positioning module status, (0:Unknown,1:Work,2:Strange,3: Not exist)	
module.record	String			Channel recording video status hexadecimal string with each bit representing the recording state of the channel, starting at 0.For example: Bit0 corresponds to the first video, and so on	0xf
payload.basic Object					
basic.key	Number			1 acc on; 0 acc off	
payload.storage Stores an array of state objects					
storage.index	Number			Disk serial number	
storage.status	Number			Disk status, 0: Unknown, 1:Recording, 2:Free, 3:Strange, 4:Disk is full	
storage.total	Number			Total storage (MB)	
storage.free	Number			The rest of storage (MB)	
payload.mileage Mileage object					
mileage.todayDay	Number			Mileage of today	
mileage.total	Number			Total mileage	
payload.obd Vehicle OBD data array object, there may be multiple objects, each object contains the following fields:					
totalMil	Number			Total mileage	
totalFuel	Number			Total fuel amount	
instanFuel	Number			Instant fuel consumption*100: L/km	
voltage	Number			Vehicle voltage*100	
rpm	Number			Engine speed RPM (Revolution(s) Per Minute)	
speed	Number			Vehicle speed: km/h	
airshed	Number			Air intake flow rate: g/s	
stressMpa	Number			Air intake pressure: kPa	
coolantsTemp	Number			Engine coolant temperature °	
airTemp	Number			Air intake temperature °	
motorLimit	Number			Engine load: %	
position	Number			Throttle position: %	
efoa	Number			Fuel tank level: %	
vin	String			VIN code	
engine	Number			Engine status: 1-ON, 0-OFF	
idle	Number			Idle status: 1-Start, 0- End	
engineOnTime	String			Engine on time	
engineOffTime	String			Engine off time	
hc	Number			Harsh cornering	
ha	Number			Harsh acceleration	
hb	Number			Harsh brake	
lowbv	Number			Battery low voltage.	

				0-No, 1-Yes.	
payload.bluetooth: Bluetooth status object					
connect	Number			Bluetooth connection status: 0: not connected 1: connected	

Json Format example:

```
{
  "action": "80003",
  "payload": {
    "deviceId": "300341", //Device name
    "location": { //Positioning information
      "mode": "1", //Positioning type, 1 is GPS
      "dtu": "2018-01-11 11:35:02", //Device time
      "direct": "59", //Direction 0~359°
      "satellites": "0", //Satellite number
      "speed": "17", //Speed, unit: KM/H
      "altitude": "21", //Altitude(meter)
      "precision": "50", //Positioning precision(meter)
      "longitude": "114.106476", //Longitude
      "latitude": "22.5767326" //Latitude
    },
    "module": { //State of the module
      "mobile": "1",
      "location": "1",
      "wifi": "1",
      "gsensor": "1",
      "record": "0xf" //Channel recording state, BIT0 corresponds to the first video, and so on
    },
    "basic": {
      "key": "1", //1 acc on; 0 acc off
    },
    "storage": [ //Stores an array of state abjects
      {
        "index": "0", //Serial number
        "status": "1", //Disk status
        "total": "60906", //Total storage(MB)
        "free": "0" //the rest of the storage(MB)
      }
    ],
    "mileage": { //mileage information
      "todayDay": "0", //On the day of the mileage(meter)
      "total": "0" //Total mileage(meter)
    }
  }
}
```

4.5 Push Device online/offline info

Request way: websocket

Request direction: ws server->ws client

Request parameter:

Json object/attribute	Type	length	Required	Description	Example value
action	number		Yes	ws message type, online/offline message Yes80005, see: ws message type table	80000
Payload object definition					
payload.deviceID attribute	string	128	Yes	Device id	99990001
payload.accessMode	Number		Yes	Device onlinenetwork type, see: device network status table	

Json example:

```
{
  "action": "80005",
```

```

    "payload": {
      "deviceId": "300342", //Device name
      "accessMode": 5 Device online network type
    }
  }

```

4.6 Device event message push (Alarm message)

Request way: websocket

Request direction: ws server->ws client

Request parameter:

Json object/attribute	Type	length	Required	Description	Example value
action	number		Yes	ws message type, event message push Yes80004, see ws message type table	80004
Payload object definition					
payload.deviceID attribute	String	64	Yes	Device id	99990001
payload.alarmID attribute	string	48	Yes	Event uuid	60D27AA2-E2E7495A-B77AB878-D58861F2
payload.payload event details object					
payload.dtu	String	24	Yes	Occurrence device time	2016-01-11 11:35:02
ec	Number	16	Yes	Type, see event code table	
St	String	24	Yes	start time	2016-01-11 11:35:02
et	String	24	Yes	end time	2016-01-11 11:35:12
det	Object		No	Alarm detailed parameters. Contains detailed alarm related parameters and alarm subtypes. You can refer to the code table.	{ "num": "sd1", "st": "0" }
payload.location Object (location message object)					
location.mode	Number	8	Yes	Location type 0—invalid 1--GPS 3--BD 4---Glonass 5--AGPS 5—base station 6—WIFI	1
location.dtu	String	24	Yes	GPS locating time	2018-01-11 11:35:02
location.direct	Number	8	Yes	direction, 0~359°	
location.speed	Number		Yes	speed, unit: KM/H	
location.altitude	String		Yes	Altitude (meter)	
location.precision	Number		Yes	Location accuracy (meter)	
location.longitude	String		Yes	Longitude	114.106476
location.latitude	String		Yes	Latitude	22.5767326
payload.obd Vehicle OBD data array object, there may be multiple objects, Please refer to the obd object in the status message					

Json example:

```

{
  "action": "80004",
  "payload": {
    "deviceId": "300342", //device ID
    "alarmID": "60D27AA2-E2E7495A-B77AB878-D58861F2", //event id
    "payload": {
      "det": { //detail
        "ch": "1" //Triggered channel, starting from 1
      },
      "dtu": "2018-01-12 09:34:20", //Device time
      "ec": "1", //Event message type
      "et": "2018-01-12 09:34:30", //end time
      "st": "2018-01-12 09:34:20"
    }
  },
  "location":

```

```
{
  {
    "mode": "1",
    "dtu": "2021-01-25 17:31:59",
    "direct": "0",
    "satellites": "0",
    "speed": "19",
    "altitude": "1833",
    "precision": "5",
    "longitude": "114.111992",
    "latitude": "22.5380726"
  },
}
```

5 Web video integration (iframe embedding method)

Iframe web page embedding supports online playback of H.264 and H.265

5.1 Device real-time video preview URL

External platforms can load the VIDEO preview HTTP URL provided by VSS through iframe in their own Web pages, and pass relevant parameters through URL to quickly complete the development of video preview function. This page supports multiple channels playing at the same time.

- ✓ Real time preview http URL address

Format example is as follows:

http://vssServerIp:9966/vss/apiPage/RealVideo.html?token=adfsdfabdddabcedfg&deviceId=20191122&chs=1_3&stream=0&wnum=4&panel=1&buffer=2000

The parameters are defined as follows:

- (1) token, required parameter, Yes the string of tokens returned after successful login through the HTTP API login interface
- (2) deviceId, required parameter, the device ID which needed to play video
- (3) chs, required parameter, channel list, each channel number is separated by an underscore”_”
- (4) stream, required parameter, Stream code, 1 represents the main-stream, 0 represents the first sub-stream
- (5) wnum, optional parameter, Default window number (Required is one of 1/4/6/9/16), when less than the number of channels, the number of channels shall prevail.
- (6) panel, optional parameter, whether to display the window layout panel. 1.display(default), 0 not display.
Note: if needs to control multiple screens by yourself, you can open one channel at a time, and make a configuration wnum=1,panel=0.
- (7) buffer, optional parameter, video buffer time, in milliseconds, default 2000. For mobile app, it is recommended to set to 0
Note: If needs to control multiple channels, you can open one channel at a time and set wnum=1,panel=0

- ✓ Real time preview https URL address

All of them are the same except that the port is different (using the default port of 443),as follows:

https://vssServerIp/vss/apiPage/RealVideo.html?token=adfsdfabdddabcedfg&deviceId=20191122&chs=1_3&stream=0

5.2 Web URL for device to listen

- ✓ Device to monitor the http URL address

The format example is as follows:

<http://vssServerIp:9966/vss/apiPage/RealListen.html?token=abcedfg&deviceId=20191122&chs=1>

The parameters are defined as follows:

- (1) Token parameter, the string of tokens returned after successful login through the HTTP API login interface
- (2) deviceId parameter, the device ID of the device which needs to play
- (3) chs parameter, channel number, starting at 1, the listener can carry only one channel number

- ✓ https URL address

- ✓ All of them are the same except that the port is different (using the default port of 443).

5.3 URL of device intercom webpage

Web intercom can only be used normally under https. VSS is a self-signed certificate by default, and the browser will prompt an access warning. It is recommended to install a certificate issued by a certificate authority.

- ✓ Device intercom https URL address

The format example is as follows:

<https://vssServerIp:36301/vss/apiPage/dialog.html?token=abcedfg&deviceId=20191122&serverIp=12.12.12.12>

The parameters are defined as follows:

- (1) **token** parameter: the token string returned after successful login through the http api login interface
- (2) **deviceId** parameter: the device ID of the device tht needs to play
- (3) **ServerIP** parameter: It is the public network address of the corresponding server (node or center), which can be IP or domain name

5.4 Web URL for Device playback video

This page supports simultaneous playback of multiple videos

- ✓ Device video replay HTTP URL address

The format example is as follows:

http://vssServerIp:9966/vss/apiPage/ReplayVideo.html?token=abcedfg&deviceId=20191122&chs=1_3&st=20200810110900&et=20200810112800&wnum=4&panel=1&speed=4&buffer=3000

The parameters are defined as follows:

- (4) token, required parameter, The string of tokens returned after successful login through the HTTP API login interface
- (5) deviceId, required parameter, the device ID of the device which needs to play
- (6) chs, required parameter, A list of channels, with each channel number separated by an underscore "_"
- (7) st, required parameter, Video start time. Format YYYYMMDDHHIISS
- (8) et, required parameter, Video end time
- (9) wnum, optional parameter, Default window number (Required is one of 1/4/6/9/16), when less than the number of channels, the number of channels shall prevail.
- (9) panel, optional parameter, Whether to display the window layout panel.1, displayed (default), 0 not displayed
- (10) Speed, fast-forwarding parameter, could be 2/4/8/16, Optional parameter, if this parameter is not carried, it will be played at normal speed
- (11) buffer, optional parameter, video buffer time, in milliseconds, default 2000. For mobile app, it is recommended to set to 0
- ✓ https URL Addresss
All of them are the same except that the port is different (using the default port of 443), as follows:

5.5 Web URL of playback alarm video clips and Server Recording clips

The small alarm video in server refers to the small video clip generated by the alarm linkage, which is finally stored on the VSS server. This webpage can only play one server video file, and does not support multiple channel play at the same time.

Server video clips: The video server actively grabs the real-time video of the device and saves it on the VSS server. The video is sliced and stored by time. This interface supports playing a certain clips of video. The video path can be obtained using the http video file search interface.

- ✓ Server video playback http URL address

Example:

<http://vssServerIp:9966/vss/apiPage/ReplayAlarmServerVideo.html?token=abcdefg&deviceId=20191122&chs=1&fpath=RSUzQSU1Q0hvd2VuJTVDRG93bmXvYWQlNUMyMDIwLTA4LExJTVDmJAxOTgwMDIINUNjaDAxXzIwMjAwODExXzAwMzAwMF8wMDUzMDJfMDEuYXNm>

The parameters are defined as follows:

- (1) The token parameter is the token string returned after successful login through the http api login interface
- (2) deviceId parameter, the device ID of the device to be played
- (3) chs parameter, channel list, each channel number is separated by underscore "_"
- (4) The fpath parameter, the full path of the server video file obtained through the http video search interface, needs to be passed in after base64 encoding

- ✓ https URL address

Except that the port is different (using the default port 443), everything else is the same.

5.6 Web URL for Server to playback of regularly downloaded video

Server video refers to the video downloaded from the device by VSS on a regular basis. This page can only play a server video file, and does not support multiple images.

- ✓ Video replay HTTP server URL

The format example is as follows:

<http://vssServerIp:9966/vss/apiPage/ReplayServerVideo.html?token=abcdefg&deviceId=20191122&chs=1&fpath=RSUzQSU1Q0hvd2VuJTVDRG93bmXvYWQlNUMyMDIwLTA4LExJTVDmJAxOTgwMDIINUNjaDAxXzIwMjAwODExXzAwMzAwMF8wMDUzMDJfMDEuYXNm>

The parameters are defined as follows: :

- (1) Token parameter, the string of tokens returned after successful login through the HTTP API login interface
- (2) deviceId parameter, the device ID of the device which needs to play
- (3) chs parameter, A list of channels, with each channel number separated by an underscore "_"
- (4) fpath parameter, the full path of the server video file obtained through the HTTP video search interface, need to be encoded by base64 and then write.

- ✓ https URL address

All of them are the same except that the port is different (using the default port of 443)

6 Web video SDK (Javascript SDK)

6.1 SDK player for fixed UI

Third-party platforms can realize online playback of various audio and video data by integrating VSS's open JavaScript player SDK. Support both H.264, and H.265

SDK api document address: http://vssServerIp:9966/vss/apiPage/api_vss.html

SDK Demo address: http://vssServerIp:9966/vss/apiPage/api_demo_player.html

6.2 SDK for player of self-defined UI

Starting from VSS v2.2, the SDK supports custom UI (while the old interface can still be used):
<http://ip:9966/vss/demo.html>

7-Video Request HTTP-FLV URL

(This chapter is no longer applicable.)

The Vss platform provides streaming media request services based on the flv over http standard protocol for device video access, and the audio is encoded by AAC. Third-party platforms can request playback through common open source players such as flvjs and vlc. Note: This solution is not recommended. This solution has a general experience, poor compatibility, and does not support H265.

1. Device real-time video request url

http://vssServerIp:33122/live?token_deviceId_channel_stream

Service port 33122, service type flag is "live"

The url needs to carry the following parameters, and each parameter is separated by underscore:

- token, Token obtained after logging in through the http web api login interface
- deviceId, Device ID
- channel, Channel number, starting from 1.
- stream, The requested stream type, 1 represents the main stream, 0 represents the sub stream

Example:

Request the first channel sub-stream audio and video stream with the device ID of 666888:

http://172.16.50.210:33122/live?a1512bf437d6c0a086d9881ba277d7fc_666888_1_0

2. Device video playback request url

http://vssServerIp:33122/replay?token_deviceId_channel_startDateTime_endDateTime

Server port 33122, service type flag is "replay"

Url needs to carry the following parameters, and each parameter is separated by underscore:

- token, Token obtained after logging in through the http web api login interface
- deviceId, Device ID
- channel, Channel number, starting from 1.
- startDateTime, Request recording start time, format: yyyyMMddHHmmss, for example 20201015123040, which means 2020-10-15 12:30:40
- endDateTime, Request recording end time, same format as above

Example:

http://172.16.50.210:33122/replay?a1512bf437d6c0a086d9881ba277d7fc_666888_1_20200810110900_20200810112800

8 Code sheet

8.1 Error code for HTTP interface

Error code	Description
10000	Operation successful
10001	Account or password error
10002	Verification code error
10003	Server error
10004	Not logged in
10005	Please fill in the mandatory parameter
10006	Device not exist
10007	Operation failed
10008	Vehicle not exist
10009	Fleet not exist
10010	Driver not exist
10011	Role not exist
10012	Account not exist
10013	Server not exist
10014	Area not exist
10015	Vehicle not in the area
10016	Account already existed
10017	Account not activated
10018	No authority
10019	Auto download plan not exist
10020	Alarm auto download plan not exist
10021	File Information of alarm auto download file not exist
10022	File information of auto download not exist
10023	Access token error
10024	Record not exist
10025	No record
10026	Illegal parameter
10027	Original password error
10028	Fail to update configuration of server
10029	Device already exist
10030	Quantity of intercom channel cannot be more than 32.
10031	Quantity of output channel cannot be more than 32.
10032	Quantity of input channel cannot be more than 32.
10033	Quantity of audio coding channel cannot be more than 32.
10034	The number of audio coding channel does not match the number of channel name.
10043	Device is offline
10036	rule does not exist
10037	The maximum device ID cannot exceed 9999
10038	license has expired
10040	Automatic download task does not exist
10041	Device IDs are inconsistent
10042	Device type does not exist
10047	unknown error
10048	Exceeds the number of licenses
10049	The file format is wrong, only png format is supported
10050	Area already in use
10051	There are users in this role
10052	There are vehicles in this fleet
10053	Role already exists
10054	Failed to upload image
10055	Card number already exists
10056	Verification code error
10057	IO alarm name cannot exceed 12
10058	Fleet name already exists
10059	There are drivers in this fleet
10060	Length cannot exceed
10061	Package name already exists
10062	There are upgrade tasks for this vehicle

10063	Task not exists
10064	Upgrade package not exists
10065	Upgrade package model name not exists
10066	Illegal upgrade package name
10067	Illegal upgrade package format
10068	Some devices still connected to this node
10069	Partial addition failed (quantity limit exceeded or same strategy exists)
10070	Name already in use
10071	License expired
10072	Abnormal operation of the database node
10073	Driver's face photo cannot exceed 1.6MB
10074	Driver's license photo cannot exceed 0.8MB
10075	You cannot delete yourself
10076	Not support this device
10077	License invalid
10078	Query time range exceeded
10079	Image exceeded upload size limitation
10080	Vehicle has been bound, please unbind it before operating
10081	Export timed out, please select export in batches
10082	Login too often
10083	Device exception
10084	Records exist under the group, cannot delete the group
10085	Invalid request method
10086	Fleet name already exists
10087	Parameter has SQL injection risk
10088	Configuration parameter value is invalid
10089	Invalid username
10090	Logged out
10091	Some configuration items updated successfully, please refresh the page to confirm
10092	Parent fleet cannot be the fleet itself
10093	Share has expired
10094	Daily traffic cannot exceed monthly traffic
10095	This fleet still has sub-fleets
10096	Name already exists
10097	Expired
10100	Invalid voice package format
10101	Voice package number or voice file name already exists
10102	Voice package does not exist
10103	Voice package issuance successful, please wait for the device to download the voice package notification
10105	Account is locked
10106	Device network may be abnormal or just offline
10107	Unknown language encoding
10108	File does not exist
10109	No device bound
10110	File upload to FTP failed
10111	Batch operation failed
10112	Voice file has corresponding issuance task, please delete the task first
10113	Test failed
10114	Already exists, cannot add again
10115	Streaming media service exception
10116	Task deadline cannot be earlier than the current date
10117	Voice name already exists
10118	Task name already exists

8.2 Device online status

Value	Description
-1	Offline
0	Unknown
1	Ethernet
2	WIFI
3	2G
4	3G

5	4G
6	5G
7	WIFI+3/4/5G, connected to mobile network agent via Wi-Fi
8	CABLE+3/4/5G, connected to mobile network agent via Ethernet

8.3 Type of data frame

Value	Description
0	Invalid
1	Main Stream (H264)
2	Virtual stream (H264)
3	Audio frame (G726 contains 2 bytes special header)
4	Serial port data frame
5	File data frame(Media data length and other information are all 0, indicating that the file transmission is complete and the video playback is complete)
6	Status data frame(Valid when played back)
7	Alarm data frame(Valid when played back)

8.4 Type of service

Value	Description
0	Invalid
1	Management
2	Gateway
3	Storage
4	Flow media

8.5 Device status

Value	Description
-1	Offline
0	Unknown
1	Online via Ethernet
2	Online via WIFI
3	Online via 2G mobile network
4	Online via 3G mobile network
5	Online via 4G mobile network
6	Online via 5G mobile network
7	Online via WIFI+3/4/5G, connected to mobile network agent via Wi-Fi
8	Online via CABLE+3/4/5G, connected to mobile network agent via CABLE

8.6 Type of file

Value	Description
0	Unknown
1	Ordinary video
2	Alarm video
3	Ordinary captured file
4	Alarm captured file
5	Update file
6	Log file
7	Configuration file
8	Black box

8.7 PTZ movement code

Value	Description
0	Unknown
1	Up
2	Down
3	Left
4	Right
5	Upper left

6	Lower left
7	Upper right
8	Lower right
9	Jump to preset position
10	Set preset position
11	Clear preset position
12	Big aperture
13	Small aperture
14	Zoom out
15	Zoom in
16	Focus near
17	Focus far
18	Auto pan
19	Start wiper
20	Stop wiper
21	Cruise on
22	Cruise off
23	Light on
24	Light off

8.8 Type of device

Value	Description
Hero-ME41-04	Howen device, Hero-ME41-04
Hero-ME40-04	Howen device, Hero-ME40-04
Hero-ME34-04	Howen device, Hero-ME34-04
Hero-ME32-04	Howen device, Hero-ME32-04
Hero-ME31-04	Howen device, Hero-ME31-04
Hero-ME34-08	Howen device, Hero-ME34-08
Hero-ME31-08	Howen device, Hero-ME31-08
Hero-MA80-08	Howen device, Hero-MA80-08
Hero-ME41-02	Howen device, Hero-ME41-02

8.9 Event code

8.9.1 Device Event Code

For details on device event codes, please refer to the unified document:

https://docs.google.com/spreadsheets/d/1zLB_oTg2bKf1Dfzwdkdfc_yps-AaVYEAtIWOCQ_QL9aI/edit?usp=sharing

Event Code		Event Name		Howen					
(H-Protocol)	(Web API)			MC series	ME series				
Main type (ec)	Sub Type (tp, dt, ...)	Main type	Sub type	MC30-01	ME40-02 V3 ME40-02 V3C	ME40-02 V8 ME40-02 V8C	ME41-02N	ME40-04N ME30-04N	ME40-04N2
st=9	st=9		9-Disk pauses to write video: 3 consecutive p	Y	Y	Y	Y	Y	Y
	st=10		10-Disk recording overwrite exception: a kind	Y	Y	Y	Y	Y	Y
	st=11		11-The disk has not recorded for a long time:	Y	Y	Y	Y	Y	Y
	st=12		12-The disk is written slowly, causing the cac	Y	Y	Y	Y	Y	Y
	st=12								
17	17	Overtime driving (fatigue)							
18	dt=1 210	fuel consumption abnormal	Refuel						
	dt=2 211		Fuel theft						
19	19	ACC Off		Y	Y	Y	Y	Y	Y
20	20	GPS module abnormal							
21	21	front panel open							
22	tp=1	Swipe card	Driver swipe card	Y	Y	Y	Y	Y	Y
	tp=2		Student swipe card	Y	Y	Y	Y	Y	Y
	tp=3		Invalid card	Y	Y	Y	Y	Y	Y
23	23	IBUTTON							
24	24	Harsh acceleration		Y	Y	Y	Y	Y	Y
25	25	Harsh braking		Y	Y	Y	Y	Y	Y
26	26	Low speed warning		Y	Y	Y	Y	Y	Y
27	27	High speed warning		Y	Y	Y	Y	Y	Y
dt=1	230		1-low voltage	Y	Y	Y	Y	Y	Y

8.9.2 VSS platform internal event code table:

Value	Description
10001	Insufficient server disk space

11001	Excessive device traffic
11003	Device offline for a long time

8.10 DMS/ADAS alarm subtype

The definition of the subtype is encapsulated in the alarm details *det* object in the DMS/ADAS alarm message.

Example of det object format:

```
{
  "tp": "70"
}
```

tp: subtype, for details, please refer to the unified document: Product function matrix table → Device event specification table

8.11 Alarm Details parameter analyze

Please refer to below tables, to analyze the parameters in Alarm Details.

Alarm service data format as below:

Message header + JSON load (alarm detail) + status data

Content	Description	
Message No.	0x1051	
Direction	MDVR →→ Server	
Link Type	signal link	
loading data	Binary coded format	
	Content	Length
	Session No.	1 byte
	length	Including terminator, if Session No.is empty, the session number field needs to add terminator, length is 1
	Session No.	N byte
	Content length	4 bytes
	Length of Alarm content description string, including terminator/ending mark	
	adopt JSON encoding rule (alarm content description)	
	Content	Field name
	Description	
	Items that must be filled in	
	Device Time	dtu
	start time	st
	End Time	et
	Event Type	ec
	Picture address	pa
	Recording address	ra
	Alarm ID	uuid
	alarm description	det
	Driver ID	drid
	Driver name	dname
	location status data (If the subscribed data is 0, it will not load the data)	
	Binary coded format	
	refer to Status Data (2.7.4, Status Data, H protocol)	

1. Video loss, motion detection, video blind, input trigger, emergency alarm, Output

Video loss, motion detection, video blind, input trigger, emergency alarm		
Content	Field name	description
channel	ch	trigger channel, starting from 1

2. Input Trigger

Input trigger		
Content	Field name	Description
Channel	ch	Triggered channel, starting from 1
Number	num	0- close. 1- Emergency/ Panic 2- F-door 3- M- door 4- B-door 5- Near light 6- Far light 9- R-Turn (right turn) 10- L-Turn (left turn) 11- Braking, 12- Reverse

		13- Reserved 1 14- F-door close 15- M-Door Close 16- B-door close 17- Talk (start the intercom) 18- Raise up 19- Airtight 20- Load 22- Custom defines 23- Safe to load 31-IBT2
--	--	--

3. High temperature, Low temperature

High temperature, low temperature		
Content	Field	Description
Type	num	0: CPU 1: HDD 2: cabin
Trigger threshold	vt	
Current value	cur	

4. Speed alarms, etc.

low speed alarm, over speed alarm, low speed warning, high speed warning, harsh acceleration, harsh braking, low temperature alarm, high temperature alarm, idle alarm, excessive high speed		
Content	Field name	description
trigger threshold	vt	conditional value, if >90 is over speed, then the trigger threshold is 90
time threshold	tt	if > 90 over 5 second is over speed, then the time threshold is 5 second
maximum value	max	Maximum speed value during the alarm
minimum value	min	Minimum speed value during the alarm
average	avg	Average speed value during the alarm
current value	cur	Current speed value when reporting
Previous second	pre	The speed value of one second before
Duration	dur	Unit: s (valid when idling, other alarms are 0)
Speed resource	spds	0—GPS speed 1—OBD speed

5. Overtime parking

Overtime parking		
Content	Field name	Description
trigger value	vt	
parking time	st	second

6. Vibration Alarm

vibration alarm / Acceleration Alarm		
Content	Field name	Description
trigger threshold	vt	When dt=5: *10 Other: *100
time threshold	tt	Anti-shake time (s): *10
maximum value	max	Maximum speed value during the alarm
minimum value	min	Minimum speed value during the alarm
current value	avg	Average speed value in a period
current value	cur	Current speed value
direction	dt	1—X direction, 2—Y direction, 3—Z direction, 4—impact, 5—tilt, 6—turn, 7—Harsh acceleration, 8—harsh braking
Previous second	pre	The speed value of one second before

7. Geofence

Electronic fencing, Electronic route		
Content	Field name	description
numbering	num	fencing or route numbering
status	st	0-enter 1-exit 2- over-speed alarm 3- over speed warning 4- low speed alarm 5- low speed warning 6- forbidden parking engine star 7-forbidden parking engine off 8-overtime stay in geofence 9-Pre-entry 10- Pre-exit
Trigger Threshold	vt	Valid when st = 2 or 4, unit: Km/h

8. Door open/close

Abnormal open/ close door		
Content	Field name	description
Channel	Ch	Triggered channel of inputs, starting from 1
numbering	num	2—front door, 3—mid door, 4—back door
status	st	0—close, 1—open

9. Storage abnormal

Storage abnormal		
Content	Field name	description
numbering	num	For example: sd1, sd2, hdd1, hdd2
status	st	0 - Missing 1 -- Broken (disk partition fatal error) 2--The log cannot be overwritten 3--Failed to write Block (EIO write error) 4--Disk failure (disk cannot be partitioned) 5--The disk cannot be mounted: the log partition cannot be mounted 6--There are too many bad blocks in disk video storage, more than 20% 7--Disk invalid block: Judge the video partition, at the beginning of formatting, or when the key information of the corresponding file is updated later 8--Disk video sampling verification failed: Failed when compare key information, after a file is finished 9--Disk pauses to write video: 3 consecutive pauses will be reported once 10--Disk recording overwrite exception: a kind of verification for false writing 11--The disk has not recorded for a long time: no recording has been written for more than 2 minutes 12--The disk is written slowly, causing the cached data to be overwritten

10. People counting

People counting		
Content	Field name	description
Front door get-on people number	Up0	The number of onbus people, <0 means invalid
Front door get-off people number	Dw0	The number of offbus people, <0 means invalid
Back door get-on people number	Up1	The number of onbus people, <0 means invalid
back door get-off people number	Dw1	The number of offbus people, <0 means invalid
Middle door get-on people number	Up2	The number of onbus people, <0 means invalid
Middle door get-off people number	Dw2	The number of offbus people, <0 means invalid
[Reserved] get-on people number	Up3	The number of onbus people, <0 means invalid
[Reserved] get-off people number	Dw3	The number of offbus people, <0 means invalid
longitude	lon	
latitude	lat	
	Pat	0-real time 1-stored and upload (data was resent after network works)
time	tm	Format: “yyyy-mm-dd hh:mm:ss”
	Va	Current onbus people number byte is valid or not (device calculated), 0-invalid , 1-valid
	Cur	the current number of people onbus

11. DMS&ADAS

Fatigue driving (DMS & ADAS)		
Content	Field name	Description
fatigue level	tp	Refer to alarm type
ID	id	
Identification name	name	

12. Fuel level abnormal alarm

fuel level abnormal alarm		
Content	Field name	Description
trigger threshold	vt	
oil tank capacity	to	unit: Liter 0-invalid / not supported
balance fuel capacity	fr	unit: Liter 0-invalid / not supported
Alarm Type	dt	1: Refuel 2: Fuel theft

13. Swipe card

Swipe Card (RFID/NFC/ Magnetic Card reader etc.)		
Content	Field name	Description

Swipe card type	tp	1-Driver, 2-Student, 3-invalid card
Swipe card info	cn	Card number, etc.
Onboard/offboard	up	1: Onboard (check-in), 2: Offboard (check out), else—invalid if for Driver swipe card: 1-login, 2-log out. Other-invalid
History	ht	1—Historical data 2—Realtime data, Else—invalid
Type	it	0—RFID; 1—IBT (i-button); 2—face recognition, 3-ibutton+Face Recognition

14. Voltage

Voltage		
Content	Field name	Description
High/low voltage	dt	1—low voltage 2—high voltage 3—Power off 4—Power on 5-Power off when moving 6—Low voltage shutdown 7—Start up
Trigger Threshold	vt	Threshold; *100
Current value	cur	Current voltage value when reporting; *100

15. Over-time Driving (Fatigue Detection)

Over-time Driving		
Content	Field name	Description
Fatigue level	de	Reserved
Lasting duration	du	Unit: second
Threshold	lt	Unit: minute
ID	Id	
Name	na	

Non-alarm event

16. Trip notification

Trip Notification		
Content	Field name	Description
Average speed	avg	
Max Speed	max	
Duration	dur	
Driver ID	drid	
Start longitude	sln	
Start latitude	slat	

17. Tire pressure

Notification of Tire pressure, (a group of tire pressures)		
Content	Field	Description
ID	id	
Temperature	temp	id=0: not exist
Pressur	pres	id=0: not exist
Group No.	gn	id=0: not exist, A1、A2、A3
Alarm data	od	id=0: exist, Format: type, raw data; type 1-tpms, 2-undefined

18. Disk detection

Disk abnormal		
Content	Field	Description
No.	num	Like: sd, sd2, hdd1, hdd2
Overwriting times	ow	

19. Load alarm

Load alarm		
Content	Field	Description
Type	dt	0--overload 1--underload 2--connection abnormality
Valid when over-loaded, under-loaded		
Trigger threshold	vt	Unit: ton
Total weight	tw	Unit: ton
Cargo weight	gw	Unit: ton
Voltage	vl	Unit: mv (current voltage of load sensor)

20. Alarm File

Notification of file generated		
Content	Field	Description
File type	ft	Refer to the file type code: File Type Code
File name	fn	

21. Alarm file in visible partition

Notification of alarm file in visible partiion generated (mp4/jpg)		
Content	Field	Description

Channel	ch	
Duration	dur	
File size	fs	
File type	ft	Refer to the file type code: File Type Code
File name	fn	

```
{"ch":"1","dur":"20","fn":"/mnt/sd2/REC-ALARM/20220811/171059_1/1_1_0_1660237859.mp4","fs":"2557615","ft":"2"}
```

22. Timer Snapshot

Timer Snapshot		
Content	Field	Description
Channel	ch	
File type	ft	Refer to the file type code: File Type Code
File name	fn	

```
{"ch":"3","fn":"/mnt/sd2/capture/20220811/1660238445073CH03T1.jpg","ft":"3"}
```

23. FTP file upload

Notification of ftp file upload		
Content	Field	Description
Channel	ch	
File type	ft	Refer to the file type code: File Type Code
File name	fn	

```
{"ch":"3","fn":"/mnt/sd2/capture/20220811/1660238445073CH03T1.jpg","ft":"3"}
```

Some data in the alarm load data is not yet implemented.

HEX example:

```
48015110bc00000020616c61726d2d32383038313130322d3030303030314539364244464230313000640000007b22646574223a7b226368223a223122
7d2c22647475223a22323031382d30392d31342031343a33313a3037222c226563223a223222c226574223a22222c227061223a22222c227374223a223
23031382d30392d31342031343a33313a3037222d0a0012090e0e1f072d00000112090e0e1f07000b00008214080071588f0800165a280500810000001f
00000103010f000000000000
```

Loading data analysis:

Session No. length: 0x20

Session No.: alarm-28081102-000001E96BDFB010

Content length: 0x00000064

Content (JSON) : {"det":{"ch":"1"},"dtu":"2018-09-14 14:31:07","ec":"2","et":"","pa":"","st":"2018-09-14 14:31:07"}

Device time: 2018-09-14 14:31:07

Start time: 2018-09-14 14:31:07

End time: Null (means triggered)

Event type: 2

Photo location: Null

Alarm description: {"ch":"1"} (means Ch1)

Location status data

Device time: 2018-09-14 14:31:07

Content: 0x002d (Means there is location info, basic status, module working status, mobile network status)

Location info

Info: 0x00

Locating type: 0x01

Time: 2018-09-14 14:31:07

Direction: 0x00

Satellite quantity: 0x0b

Speed: 0x0000

Altitude: 0x1482

Location accuracy: 0x0008

Longitude degree: 0x71

Longitude minute: 0x00088f58

Latitude degree: 0x16

Latitude minute: 0x0005285a

Basic Status

Identifier bit 1: 0x81 (Means ignition key valid, right front door open)

Identifier bit 2: 0x00

Reserve: 0x0000

Module working status

Identifier bit: 0x001f (Means there exists: mobile network data, location module data, Wi-Fi module data, G-sensor data, recording status data)

Mobile network: 0x00

Location module: 0x01

WIFI module: 0x03

G-sensor: 0x01

Recording status: 0x000f

Mobile network status

Identifier bit 1: 0x00

Signal strength: 0x00

Network type: 0x00

Reserved: 0x0000

8.12 ws Message type table

Value	Description
-------	-------------

80000	The client will request login verification after successful connection, and automatically start subscription after successful connection
80001	Subscribe to real-time messages
80003	The server pushes device status information to the client
80004	The server pushes device alarm event to the client
80005	The server pushes device online/offline information to the client
80009	The client sends heartbeat information to server at regular intervals